

Fréchet Regression and Functional Data

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1. Introduction

This thesis undertakes the task of comparing different approaches to the problem of density regression. Specifically, we will juxtapose the novel Fréchet regression method against a particular method from Functional Data Analysis (fda) literature. Within the scope of this thesis, the term Fréchet regression is confined to the global regression context.

In the fda literature, various methods have been adapted to handle density functions as responses within a regression framework. Density data have found applications in a multitude of fields, such as income distributions (Hildenbrand et al., 1999; Kneip and Utikal, 2001) and financial markets (Kokoszka, Miao, et al., 2019). However, the space of density functions is inherently not a linear space, thereby violating key assumptions in fda. This non-linearity makes it challenging to ensure that the outcomes of standard procedures remain in the density space.

Our primary focus is the transformation method as delineated by Petersen and Müller (2016), which employs a log quantile density (LQD) transformation. This approach has been slightly modified by Kokoszka, Miao, et al. (2019) and ensures that the results stay within the density space by performing standard fda methods in a transformed space and then inverting this transformation. A comprehensive overview of fda can be found in Wang et al. (2016), and an in-depth discussion on the modeling of density functions is presented in Petersen, Zhang, et al. (2022).

In contrast, another emerging avenue of research is object-oriented data analysis, which offers a more generalized perspective on functional data. Here, functional data serve as a special case within a broader category of random objects. Petersen and Müller (2019) have delved into this approach and proposed a regression method where the response variable is assumed to be in a metric space. This flexibility makes it a more direct and adaptable alternative to the transformation methods and other adaptations of fda, especially since it doesn't rely on the "extrinsic" properties of the Hilbert space of which the density space is a subset.

To empirically compare these two frameworks, a simulation study will be conducted. This study will assess the efficacy of each method in scenarios where density samples are directly observed and in more realistic settings where these samples are estimated. The remainder of this thesis is organized as follows: Section 2 offers a mathematical foundation for fda and the Fréchet regression framework, Section 3 explains basic fda concepts. Section

4 discusses density regression and the inadequacy of directly applying fda methods on the space of densities, along with the transformation approach. Section 5 explains the Fréchet method, with a special focus on our regression context in the Wasserstein space of distributions. Section 6 illustrates the methods, and compares them in a simulation study. Section 7 concludes.

The code to this thesis can be found at https://github.com/mirkimirk/thesis_frechet_fda.

2. Mathematical Background

This section provides mathematical background for fda and the Wasserstein space. Basic concepts and notation are established that will be used throughout this thesis.

2.1. Hilbert Space Theory and $\mathcal{L}^2$ Space

This section will introduce basic mathematical concepts required to explain basic fda methods. In later sections, these will be used for derivations and proofs. It is mostly based on Hsing and Eubank (2015, Chapter 2).

2.1.1. Separable Hilbert Spaces

For many methods in fda to work and be theoretically justified, additional structure on the set of functions under consideration is assumed. This structure is most often the $\mathcal{L}^2$ space of square integrable functions. This space is a separable Hilbert space. To define $\mathcal{L}^2$ and understand its properties as a separable Hilbert space, we first need the following definitions:

Definition 2.1 (Inner Product and Inner Product Space)

Let V be a real vector space. A function $\langle \cdot, \cdot \rangle_V$ that maps any two vectors $s, t \in V$ to a real number $\langle s, t \rangle_V$ is called an *inner product* if it satisfies the following properties:

1. **Symmetry:** $\langle s, t \rangle_V = \langle t, s \rangle_V$ for all $s, t \in V$.
2. **Linearity in First Argument:** $\langle as + bt, w \rangle_V = a \langle s, w \rangle_V + b \langle t, w \rangle_V$ for all $s, t, w \in V$ and $a, b \in \mathbb{R}$.

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3. Positive Definiteness: $\langle t, t \rangle_V \geq 0$ for all $t \in V$, and $\langle t, t \rangle_V = 0$ if and only if $t = 0$.

A real vector space V equipped with such an inner product is called an *inner product space*.

To generalize all the notions of interest for fda from Euclidean space to possibly infinite dimensional spaces, we need the following space:

Definition 2.2 (Hilbert Space)

Let $\mathcal{H}$ be an inner product space, and denote by $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ the inner product of this space. Then $\|x\|_{\mathcal{H}} = \sqrt{\langle x, x \rangle_{\mathcal{H}}}$ is the norm induced by the inner product, and $d_{\mathcal{H}}(a, b) = \|a - b\|_{\mathcal{H}}$ is the distance induced by the norm. If $\mathcal{H}$ is complete with respect to this metric, i.e., if every Cauchy sequence of elements in $\mathcal{H}$ converges in $\mathcal{H}$, then $\mathcal{H}$ is called a *Hilbert space*.

This concept helps generalizing many useful notions from the finite-dimensional Euclidean space in multivariate analysis to infinite-dimensional function spaces. The norm $\|\cdot\|_{\mathcal{H}}$ measures the size of an element in a Hilbert space $\mathcal{H}$ and makes it comparable with others. It also induces a metric via $d_{\mathcal{H}}(a, b) = \|a - b\|_{\mathcal{H}}$ to measure the distance between two elements in $\mathcal{H}$. The concept of inner product makes it possible to evaluate the angle between two vectors in $\mathcal{H}$, which is useful for orthogonal decompositions of this space. Especially the latter will be very important for dimensionality reduction.

To get closer to the goal of dimensionality reduction, we first need some more definitions to explain the kind of lower-dimensional subspaces we are interested in, and how the concept of orthogonality plays into that.

Definition 2.3 (Closed Span)

Let A be a subset of elements in a Hilbert space $\mathcal{H}$. The *closed span* of these elements is the closure of their linear span with respect to the metric induced by the norm, and will be denoted $\overline{\text{Span}(A)}$.

Definition 2.4 (Orthonormal Sequence and Orthonormal Basis)

A sequence $\{e_n\}$ in a Hilbert space $\mathcal{H}$ is called an *orthonormal sequence* if it satisfies:

$$\langle e_i, e_j \rangle_{\mathcal{H}} = \delta_{ij}$$

where δ_{ij} is the Kronecker delta.

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If $\overline{\text{Span}(\{e_n\})} = \mathcal{H}$, then $\{e_n\}$ is called an *orthonormal basis* of $\mathcal{H}$.

Theorem 2.4.1 (Separability and Orthonormal Basis in Hilbert Spaces)

A Hilbert space $\mathcal{H}$ is *separable* — i.e., contains a countable dense subset — if and only if it has an orthonormal basis.

Later we will be approximating elements of a Hilbert space by a combination of a finite number of its elements. So an orthonormal basis assures us, that there is this dense subset that can get arbitrarily close to any element in the space. To find the most efficient subspace (in the sense of giving the best approximation of any element for a given number of dimensions of the subspace), we want to exploit some more structure of the space $\mathcal{L}^2$ that we will be interested in. This structure is given by its separability and Hilbert space properties.

In a separable Hilbert space, the existence of a countable orthonormal basis allows us to approximate any element f in the space arbitrarily closely by finite linear combinations of the basis elements. This foundational idea is critical for methods like Functional Principal Component Analysis, where we approximate complex functions by projecting them onto a finite subset of this basis. This will be explained further in Section 3.2.

2.1.2. The Space $\mathcal{L}^2([a, b])$

In the context of fda, it is assumed that the functional data live in the space of square integrable functions $\mathcal{L}^2([a, b])$, with common support $[a, b]$.

Definition 2.5 (Space of square integrable functions $\mathcal{L}^2$)

Let $\mathcal{A} \subseteq \mathbb{R}$ be compact. The space $\mathcal{L}^2(\mathcal{A})$ consists of all functions $f : \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\int_{\mathcal{A}} (f(x))^2 dx < \infty$$

Its norm, inner product, and distance are denoted by $\|\cdot\|_2$, $\langle \cdot, \cdot \rangle_2$, and $d_2(\cdot, \cdot)$, respectively.

The concepts introduced here will be needed for an explanation of fda methods in Section 3. Before that, we will turn to introducing basic statistical notions in function spaces and the Wasserstein geometry that will be important later.

2.2. Statistics in Functional Spaces

Here, we will introduce distribution functions, the concept of Fréchet means as generalizations of the usual expected values for arbitrary metric spaces, and the Wasserstein geometry that we will deal with in the Fréchet regression context.

2.2.1. Preliminaries and Definitions

A very basic operation we will use throughout this work is switching between different functions that characterize a distribution. We will deal with the following four important classes of functions that characterize a distribution:

Definition 2.6 (pdfs, cdfs, qfs, and qdfs)

Let $\mathbb{R}^+$ be the set of nonnegative reals.

1. The *probability density function* (pdf) is a function $f : \mathbb{R} \rightarrow \mathbb{R}^+$ such that $\int_{\mathbb{R}} f(x) dx = 1$ and $f(x) \geq 0$ for all $x \in \mathbb{R}$.
2. The *cumulative distribution function* (cdf) is given by $F(x) := \int_{-\infty}^x f(t) dt$, where f is the pdf. Note that F is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
3. The inverse of F is called the *quantile function* (qf), and is denoted by Q . It is given by $Q(u) := F^{-1}(u) = \inf\{x \in \mathbb{R} \mid F(x) \geq u\}$ for $u \in [0, 1]$.
4. The derivative of $Q(u)$ w.r.t. u is called the *quantile density function* (qdf), and is denoted by q . It is given by $q(u) := \frac{d}{du} Q(u)$ for $u \in [0, 1]$.

For clarity, we will use the arguments x and z for pdfs and cdfs, and the arguments u and v for qfs and qdfs. Since we will later only look at densities from continuous distributions on their support (i.e., where $f > 0$), we will in the following assume that F is strictly increasing and thus its inverse Q to be well-defined. The following relation between the pdf and qdf will be useful later when computing transformations of the density functions (cf. Jones, 1992):

Lemma 2.7

Let f be a pdf, F the corresponding cdf, Q the corresponding qf, and q the corresponding qdf. Then it holds that

$$q(u) = \frac{1}{f(Q(u))}, \tag{1}$$

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and

$$f(x) = \frac{1}{q(F(x))}, \quad (2)$$

Proof. We will show that (1) holds, (2) follows analogously. Note that $q = (F^{-1})'$, so we can use the inverse function rule to characterize q with the inverse $Q = F^{-1}$ and derivative $f = F'$ of F :

$$(F^{-1})'(u) = \frac{1}{F'(F^{-1}(u))} = \frac{1}{f(Q(u))} \quad (3)$$

■

Because of the relation in (1), the qdf is sometimes called the "sparsity function" (cf. Tukey, 1965).

The distributions that are of interest to us are given in the following definition:

Definition 2.8

Denote by $\mathcal{G}$ the space of absolutely continuous probability distributions G supported on their own compact interval $[a_i, b_i]$ for some $a_i, b_i \in \mathbb{R}$ with $a_i < b_i$, and having a finite second moment. A distribution G is in $\mathcal{G}$ if it admits a density function f such that

$$\int_{a_i}^{b_i} x^2 dG(x) = \int_{a_i}^{b_i} x^2 f(x) dx < \infty.$$

Note that the distributions need not share any point in their support.

Lastly, we denote by $\mathcal{F}$ the density-valued stochastic process producing our sample.

2.2.2. Fréchet Mean and Variance

Let (Ω, d) be a metric space and Y a random variable that takes values in Ω . We define the Fréchet mean by

$$Y_{\oplus} = \arg \min_{\omega \in \Omega} \mathbb{E} [d^2(Y, \omega)] \quad (4)$$

This concept generalizes the estimation of mean values to any metric space. If d is the Euclidean distance d_E , this mean reduces to the usual mean $\mathbb{E}(Y)$. Accordingly, we define Fréchet variance as

$$V_{\oplus} = \mathbb{E} [d^2(Y, Y_{\oplus})] \quad (5)$$

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In Section 5.1, this notion will be generalized to that of a conditional Fréchet mean, to make use in a regression context and define Fréchet regression. The metric we will be using throughout is the 2-Wasserstein distance defined in the following section.

2.2.3. Wasserstein Distance and Its Properties

The metric space that we are interested in is the space $\mathcal{G}$ equipped with the Wasserstein-2 metric, defined in the following:

Definition 2.9 (2-Wasserstein Distance)

For $f, g \in \mathcal{G}$, consider the collection $\mathcal{M}_{f,g}$ of maps M such that if $U \sim f$ and $M \in \mathcal{M}_{f,g}$, then $M(U) \sim g$. The 2-Wasserstein distance between f and g is defined as:

$$d_W(f, g) = \left(\inf_{M \in \mathcal{M}_{f,g}} \int (M(u) - u)^2 f(u) du \right)^{1/2}$$

Before turning to the consequences for the Fréchet mean and its computation in the next section, we discuss some properties and characterizations of this distance.

While in general the Wasserstein distance is not simple to evaluate, in our case it can be shown that it is equal to the quantile distance, given by

$$d_Q(f, g)^2 := d_Q(Q_f, Q_g)^2 = \int_0^1 (Q_f(t) - Q_g(t))^2 dt,$$

where Q_f, Q_g are the quantile functions corresponding to the densities $f, g \in \mathcal{G}$ space.

Lemma 2.10

$$d_Q = d_W$$

See Petersen, Liu, et al. (2021) and the supplement of Petersen and Müller (2016) for proofs.

Setting the distance function to be d_W in (4) yields the Wasserstein–Fréchet mean $f_{\oplus}$ (cf. Petersen, Liu, et al., 2021). The following lemma shows that it can be estimated by simple averaging in the space of quantile density functions:

Lemma 2.11

The Wasserstein–Fréchet mean estimator is given by

$$\hat{f}_\oplus(x) = \frac{1}{\hat{q}_\oplus(\hat{F}_\oplus(x))}, \quad (6)$$

with $\hat{q}_\oplus = \frac{1}{n} \sum_{i=1}^n q_i$, and $\hat{F}_\oplus$ the cdf corresponding to $\hat{q}_\oplus$.

Proof. By Lemma 2.10, we can substitute the quantile distance for the Wasserstein distance in the computation of the Fréchet mean. We set $Q_\oplus(t) = \mathbb{E}[Q(t)]$ because the pointwise mean $\mathbb{E}[Q(t)]$ is the minimizer of

$$\begin{aligned} \mathbb{E}[d_W^2(f_i, f_\oplus)] &= \mathbb{E} \left[\int_0^1 (F_i^{-1}(t) - F_\oplus^{-1}(t))^2 dt \right] \\ &= \mathbb{E} \left[\int_0^1 (Q_i(t) - Q_\oplus(t))^2 dt \right], \end{aligned} \quad (7)$$

see Panaretos and Zemel (2020, Chapter 3.1.4). To compute the corresponding density function, we use (2) to get $f_\oplus(x) = \frac{1}{q_\oplus(F_\oplus(x))}$, which shows that it suffices to estimate $q_\oplus$. Since $q_\oplus = \frac{dQ_\oplus}{dt}$, we can pass the differentiation inside the expectation to see $q_\oplus = \mathbb{E} \left[\frac{dQ}{dt} \right] = \mathbb{E}[q]$, which by the analogy principle suggests averaging the observed (or previously estimated) quantile densities q_i to obtain an estimator $\hat{q}_\oplus$ for $q_\oplus$. ■

3. Functional Data Analysis

This section will introduce basic fda concepts and methods used throughout this thesis. Proofs will be omitted and can be found in the cited literature.

3.1. Covariance Operator and Covariance Functions

Here we will briefly talk about properties of the covariance operator. It is assumed that the functions under consideration here are realizations of an $\mathcal{L}^2$ stochastic process, as defined in Ash (1975, Chapter 1.1.2). We denote the stochastic process as $\{X(t) : t \in T\}$. The section is mostly based on Horváth and Kokoszka (2012, Chapter 2).

3.1.1. Definition

We will use the following definition of the covariance operator:

Definition 3.1 (Covariance Operator)

Let X be a square-integrable random curve in $\mathcal{L}^2([a, b])$, i.e.,

$$\mathbb{E} \left[\int_a^b X^2(t) dt \right] < \infty.$$

and let $\mathbb{E}[X] = \mu$. Then, the covariance operator Γ is defined for any $y \in \mathcal{L}^2([a, b])$ as

$$\Gamma(y) = \mathbb{E} [\langle (X - \mu), y \rangle_2 (X - \mu)].$$

The conditions in this definition are functional analogues of the usual conditions of finite first and second moments. This operator's properties and estimation will be further explained in the following section after introducing some important theorems for the computation and interpretation of the covariance operator.

3.1.2. Important Theorems

Here we will introduce important theorems that characterize the covariance operator and will help us obtain the functional principal components with which we will span an optimal (in the sense that was mentioned in the previous section) lower dimensional subspace.

Theorem 3.1.1 (Riesz Representation Theorem)

For every continuous linear functional Γ on a Hilbert space $\mathcal{H}$, there exists a unique $h_0 \in \mathcal{H}$ such that

$$\Gamma(h) = \langle h, h_0 \rangle_{\mathcal{H}}, \quad \forall h \in \mathcal{H}.$$

See Conway (2007, Chapter 1, §3). This theorem is crucial for interpreting the covariance operator as it can be considered an integral operator that can be represented as an inner product:

$$(\Gamma(f))(s) = \int_T \gamma(s, t) f(t) dt = \langle \gamma(s, \cdot), f \rangle, \quad (8)$$

where $\gamma(s, t) = \text{Cov}(X(s), X(t)) = \mathbb{E}[(X(s) - \mathbb{E}[X(s)])(X(t) - \mathbb{E}[X(t)])]$. It also aids in finding solutions to the Fréchet optimization problem in Wasserstein space later.

Theorem 3.1.2 (Mercer's Theorem)

Given a symmetric, non-negative definite kernel $\gamma(x, y)$ defined on a product space $A \times A$,

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where A is a subset of $\mathbb{R}$, the kernel can be represented as

$$\gamma(x, y) = \sum_{n=1}^{\infty} \lambda_n \phi_n(x) \phi_n(y).$$

Theorem 3.1.3 (Karhunen-Loève Decomposition)

Let $X(t)$ be a zero-mean square-integrable stochastic process with a covariance function $\gamma(s, t)$. Then $X(t)$ can be represented as

$$X(t) = \sum_{n=1}^{\infty} \sqrt{\lambda_n} Z_n \phi_n(t),$$

where Z_n are uncorrelated zero-mean random variables, and $\phi_n(t)$ and λ_n are the eigenfunctions and eigenvalues of $\gamma(s, t)$, respectively.

Mercer's Theorem enables the calculation of eigenfunctions and eigenvalues of the covariance operator, providing an optimal orthonormal basis that captures the most important directions of variation. When truncated to a finite-dimensional basis, this offers the best possible representation of elements in the Hilbert space. Karhunen-Loève Decomposition further enhances this by offering a powerful tool for dimensionality reduction and noise filtering in stochastic processes.

3.2. Functional Principal Component Analysis

The most popular method for describing structure in our functional data is Functional Principal Component analysis (FPCA). This is an analogue to Principal Component Analysis (PCA) from multivariate statistics in the case of infinite dimensions. It builds on the Karhunen-Loève decomposition (described in Section 3.1.2) to recover functions that describe the main modes of variation, in descending order.

We refer to the textbooks Ramsay and Silverman (2005) and Kokoszka and Reimherr (2017) for any computational details and visualization methods used in this thesis.

4. Density Regression

In this chapter we are going to examine the case of density regression, i.e., a regression model with densities as responses.

4.1. Approaches

Here, we will briefly outline different interpretations of the density space, and the approaches that follow, to motivate our choice of the transformation method from Petersen and Müller (2016) and Kokoszka, Miao, et al. (2019).

4.1.1. Subset of Hilbert Space

If one additionally assumes that $\mathcal{G} \subseteq L^2(\mathbb{R})$, one can choose to interpret the densities as elements of the Hilbert space $\mathcal{L}^2$ and directly apply fda methods on the densities (see e.g. Kneip and Utikal, 2001). Then one could handle the regression problem by applying the functional response regression method first described by Faraway (1997).

However, as $\mathcal{G}$ is not a linear subspace of $\mathcal{L}^2$, it is not guaranteed that we yield results within $\mathcal{G}$ — it only means we will yield results within $\mathcal{L}^2$, so our estimates, modes of variation etc. are not guaranteed to be themselves densities. Alternative characterizations of the space $\mathcal{G}$ have been explored to tackle this problem, for example the interpretation of elements in $\mathcal{G}$ as so-called compositional data, or as elements of the Wasserstein-2 space. The latter interpretation will be explored in Section 5 as this general approach to regression can easily incorporate the geometry of a given space that is implied by its metric (in this case, the Wasserstein-2 metric d_W .)

Another approach is the general characterization of $\mathcal{G}$ as a nonlinear space with certain constraints that can be done away with through a preliminary transformation to a linear representation space. The compositional data approach can be seen as a special case of this general transformation approach described in Petersen and Müller (2016). The special case of this approach that we will explore in the following section is the LQD transformation.

4.1.2. LQD Transformation

Instead of the approach above, one can recognize the nonlinear character of the density space and accomodate the use of fda methods such that it is respected. Several authors have proposed a prior transformation of the densities into a linear space (e.g. Hron et al., 2016; Petersen and Müller, 2016) where one can then work with the usual fda methods

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on the transformed densities, and then later transform the results back to density space for interpretation and visualization purposes.

Following the argument in Kokoszka, Miao, et al. (2019), we use the method first outlined in Petersen and Müller (2016) and use the LQD transformation $\psi : \mathcal{G} \rightarrow \mathcal{L}^2([0, 1])$, defined by

$$\psi(f)(u) = \log q(u) = -\log f(Q(u)), \quad u \in [0, 1], \quad (9)$$

to transform the pdfs from the constrained space $\mathcal{G}$ into the "unconstrained" space $\mathcal{L}^2([0, 1])$.¹ The intuition is that, from the functions given in Definition 2.6, the qf and qdf have the least constraints — only requiring that $q = Q' \geq 0$. Thus by working with the logarithm of the qdf, one deals with a function that has no more constraints (cf. Kokoszka, Miao, et al., 2019).

However, since the data generating process (DGP) that we will use in our simulation study produces densities that are on different supports, we violate the assumption of common compact support of the densities in the density sample that Petersen and Müller (2016) need.² This leads to problems since the information about location of support is lost when transforming a pdf to a qdf.³ Hence we use the slightly modified transformation and inverse transformation described by Kokoszka, Miao, et al. (2019). In this paper a very general transformation method is described for the case of different supports. Denote by $\mathcal{A}_i$ the support of the density f_i . For generality, we assume

$$\exists i, j : \mathcal{A}_i \cap \mathcal{A}_j = \emptyset,$$

i.e., we do not want to assume a common "anchor" point in the supports of the f_i . So we use the following most general transformation proposed by the authors:

$$T : \mathcal{G} \rightarrow \mathcal{L}^2([0, 1]) \times \mathcal{A}, \text{ given by } T(f) = (\psi(f), Q(u^*)), \quad (10)$$

where $\mathcal{A}$ is the support of density f , Q is the corresponding quantile function, and $u^* \in [0, 1]$ is a fixed percentile level for which we save the value of the support of f . Since

¹We extend our usage of the argument u to this function because this transformed density is by definition on the same support as are qfs and qdfs.

²We also violate this assumption because of our method of dealing with numerical issues, see Appendix A.

³Slightly paraphrasing Kokoszka, Miao, et al. (2019), the transformation from f with quantile density q to its LQD is not invertible, since for any constant c , the quantile density of $f(\cdot - c)$, the density f shifted by c , is also q .

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we deal with a compact support, we can⁴ choose the boundary value 0, which is most intuitive: we save the left bound of the support of f , and for the inverse transformation we add that bound when computing the quantile function Q from the LQD. The (forward) transformation method is described in Algorithm 1, and the inverse transformation in Algorithm 2.

Algorithm 1 Forward transformation

Require: Data pairs $(x_l, f(x_l)), l = 0, \dots, L$

Ensure: Data pairs $(u_l, \psi(f)(u_l)), l = 0, \dots, L$ and c

- 1: **for** $l = 1, \dots, L$ **do**
 - 2: Compute $F(x_l) = \int_{x_0}^{x_l} f(x) dx$ by numerical integration of the pairs $(x_l, f(x_l))$
 - 3: Define $u_l = F(x_l)$ so that $Q(u_l) = x_l$
 - 4: Compute $\psi(f)(u_l) = -\ln(f(x_l))$
 - 5: Find j such that $x_j = 0$ and set $c = Q(0)$
 - 6: **end for**
-

Algorithm 2 Backward transformation

Require: Data pairs $(u_l, g(u_l)), l = 0, \dots, L$ and c

Ensure: Data pairs $(x_l, f(x_l)), l = 0, \dots, L$

- 1: **for** $l = 1, \dots, L$ **do**
 - 2: Compute $Q(u_l) = c + \int_{u^*}^{u_l} \exp\{g(s)\} ds$ by numerical integration of the pairs $(u_l, \exp\{g(u_l)\})$
 - 3: Define $x_l = Q(u_l)$ so that $F(x_l) = u_l$
 - 4: Compute $f(x_l) = \exp\{-g(u_l)\}$
 - 5: **end for**
-

Here, g is some function from the transform space $\mathcal{L}^2([0, 1])$, resulting from some calculations we performed with the transformed densities. For the computation of the inverse, we need to supply a value of c for this function g . In our application we will deal with two kinds of functions g : estimated LQDs from our functional regression for some grid of predictor values; and a mean LQD computed from the LQD sample. For the former, we estimated c by defining a function $c(p)$, which assigns each c to the predictor value p with which it was observed, and calculates $\hat{c}$ via linear interpolation. For the mean LQD, we computed the mean of the c values in the sample and used that as an estimator. These $\hat{c}$ are supposed to estimate the left bound of the support of the resulting density when performing the inverse transform T^{-1} described in Algorithm 2.

A word about the application of fda methods in the transformation approach: the concepts introduced in Section 3.2 are always applied to objects in the transformed space.

⁴Kokoszka, Miao, et al. (2019) excluded 0 and 1 as possible values for u^* . This could have been motivated by having a possibly unbounded support of the pdfs which could yield $Q(0)$ or $Q(1)$ to be infinity, so useless for the transformation method. But this must have been a typo, since they also only considered densities with bounded support, as defined in their appendix.

The resulting functions are then transformed back to the density space. So for example, the (truncated) Karhunen-Loève representation of the densities is the translated representation of the transformed functions. Similarly for the modes of variation.

As an illustration for the difference in quality of the LQD method to the naive method, Figure 1 shows the truncated Karhunen-Loève representations of a sample density with both methods. One can clearly see that the representations based on the LQD method produce much better fitting curves than the naive method. Further illustrations, as well as the description of the DGP behind this density, are given in Appendix C.1.

In the following section, we will explain a functional regression model based on the LQD method.

4.2. Functional Regression Model

The transformation approach employed by Talská et al. (2018) can be considered a special case of the general transformation approach explained by Petersen and Müller (2016) (cf. Petersen, Zhang, et al., 2022). So we follow their modelling of a functional response model for our LQD transformation method.

4.2.1. Construction of Model

In the following, n denotes the sample size and r denotes the number of scalar predictors p . Our original regression model of interest is

$$f_i(x) = \beta_0(x) + \sum_{j=1}^r p_{ij}\beta_j(x) + \varepsilon_i(x), \quad i = 1, \dots, n, \quad (11)$$

where f_i is the i -th observed density, p_{ij} the i -th observed value of the j -th predictor, ε_i is an i.i.d. functional random error in $\mathcal{L}^2([0, 1])$ with mean zero, and lastly β_j is the j -th functional regression coefficient of the j -th predictor p_j . This can be written simpler in matrix notation as

$$\mathbf{f}(x) = \mathbf{P}\boldsymbol{\beta}(x) + \boldsymbol{\varepsilon}(x). \quad (12)$$

where $\mathbf{f}$ is the n -dimensional vector of observed pdfs, $\mathbf{P}$ is an $n \times (r+1)$ matrix containing the scalar predictor values p_{ij} (and the first column containing only ones), $\boldsymbol{\beta}$ is the $(r+1)$ -dimensional vector of functional regression coefficients, and $\boldsymbol{\varepsilon}$ is the n -dimensional vector of functional random errors.

4. Density Regression

Applying the LQD transformation (9) to (11) yields

$$\psi(f_i)(u) = \psi(\beta_0)(u) + \sum_{j=1}^r p_{ij} \psi(\beta_j)(u) + \psi(\varepsilon_i)(u), \quad i = 1, \dots, n. \quad (13)$$

Defining $\psi(\boldsymbol{\beta})$ as the $(r+1)$ -dimensional vector of transformed regression coefficients, we estimate $\psi(\boldsymbol{\beta})$ by minimizing the sum of squared errors:

$$\psi(\hat{\boldsymbol{\beta}})(u) := \arg \min_{\beta_0, \dots, \beta_r} \sum_{i=1}^n \|\psi(\varepsilon_i)(u)\|_2^2 = \int \sum_{i=1}^n \psi(\varepsilon_i)^2(u) du = \sum_{i=1}^n \left(\psi(f_i)(u) - \sum_{j=0}^r p_{ij} \beta_j(u) \right)^2, \quad (14)$$

where $p_{i0} = 1 \forall i$, and the integral and summation operations have been exchanged in the first equality because they are both linear operators (cf. Ramsay and Silverman, 2005, Chapter 13.4; Kokoszka and Reimherr, 2017, Chapter 5.1).

4.2.2. Computational Details

Since there are no particular restrictions on the functions in the transformed space, straightforward pointwise minimization can be performed by calculating

$$\psi(\hat{\boldsymbol{\beta}}) = (\mathbf{P}^T \mathbf{P})^{-1} \mathbf{P}^T \psi(\mathbf{f}). \quad (15)$$

Practically, we observed $\psi(\mathbf{f})$ on a dense grid of evaluation points and accordingly computed a discretized version of $\psi(\hat{\boldsymbol{\beta}})$ with the above method. Using linear interpolation between these values, we obtained an approximation of $\psi(\hat{\boldsymbol{\beta}})$.

We obtain the estimated pdf for any particular predictor values $p_1, \dots, p_r$ by calculating

$$\psi(\hat{f})(u) = \psi(\hat{\beta}_0)(u) + \sum_{j=1}^r p_j \psi(\hat{\beta}_j)(u), \quad (16)$$

and applying the inverse transformation.

Using $\psi(\hat{\boldsymbol{\beta}})$, we can obtain fitted values and predictions in the transformed space. These results can be translated back to density space via inverse transformation for interpretation.

We will present this regression method in section 7 where we will compare it against the Fréchet regression method. To that end, we will explain the Fréchet regression model from Petersen and Müller (2019) in the following section.

5. Fréchet Regression

This section will give an introduction to the concept of Fréchet regression and some background.

5.1. Conditional Fréchet Mean

The authors extend the notion of Fréchet mean given in Section 2.2.2 to that of a conditional Fréchet mean, with a corresponding conditional Fréchet variance:

$$m_{\oplus}(x) = \arg \min_{\omega \in \Omega} M_{\oplus}(\omega, x) \quad (17)$$

$$M_{\oplus}(\cdot, x) = \mathbb{E} \left[d^2(Y, \cdot) \mid X = x \right] \quad (18)$$

This is done by characterizing the well-known linear regression model in terms of a sequence of weighted means. For the Euclidean case, this characterization is given in the following lemma:

Lemma 5.1

Let $X \in \mathbb{R}^r$, $m(x) := \mathbb{E}(Y|X = x)$, and let μ be the mean and Σ the covariance matrix of X . Define

$$s(z, x) := 1 + (z - \mu)^T \Sigma^{-1} (x - \mu)$$

Then $m(x)$ is the value that minimizes

$$\mathbb{E} \left[s(X, x) d_E^2(Y, y) \right]$$

over $y \in \mathbb{R}$, i.e.,

$$m(x) = \arg \min_{y \in \mathbb{R}} \mathbb{E} \left[s(X, x) d_E^2(Y, y) \right] \quad (19)$$

Proof. First, we express $m(x)$ as

$$m(x) = \beta_0^* + (\beta_1^*)^T (x - \mu)$$

5. Fréchet Regression

and introduce the estimated model $\hat{m}(x)$ as

$$\hat{m}(x) = \beta_0 + \beta_1^T(x - \mu)$$

We want to characterize β_0^* and β_1^* as the optimal solutions to the minimization of the quadratic risk

$$\mathbb{E}_X \left[d_E^2(m(x), \hat{m}(x)) \right]$$

where $\mathbb{E}_X$ denotes the expectation with respect to X .

Expanding this objective function, we get

$$\mathbb{E}_X \left[d_E^2(m(x), \hat{m}(x)) \right] = \mathbb{E}_X \left[\left(\mathbb{E}(Y|X = x) - (\beta_0 + \beta_1^T(x - \mu)) \right)^2 \right]$$

To find the values of β_0 and β_1 that minimize the objective function, we take the derivatives and set them to zero. First, considering the derivative with respect to β_0 , we have:

$$\begin{aligned} -2\mathbb{E}_X \left[\mathbb{E}(Y|X = x) - \beta_0 - \beta_1^T(x - \mu) \right] &= 0 \\ \iff \mathbb{E}_X \left[\mathbb{E}(Y|X = x) \right] &= \beta_0 + \beta_1^T(\mathbb{E}_X[x] - \mu) \\ \iff \mathbb{E}(Y) &= \beta_0 + \beta_1^T \cdot 0, \end{aligned}$$

where in the last equality a continuous version of the law of total probability was used to write $\mathbb{E}_X[\mathbb{E}(Y|X = x)] = \mathbb{E}(Y)$, and $\mathbb{E}_X[x] = \mu$. Now, considering the derivative with respect to β_1 , we have:

$$\begin{aligned} \mathbb{E}_X \left[\left(\mathbb{E}(Y|X = x) - \beta_0 - \beta_1^T(x - \mu) \right) (x - \mu)^T \right] &= 0 \\ \iff \mathbb{E}_X \left[\mathbb{E}(Y|X = x)(x - \mu)^T \right] &= \beta_0 \mathbb{E}_X \left[(x - \mu)^T \right] + \beta_1^T \mathbb{E}_X \left[(x - \mu)(x - \mu)^T \right] \\ \iff \mathbb{E}_X \left[\mathbb{E}(Y(x - \mu)^T | X = x) \right] &= \beta_0 \cdot 0 + \beta_1^T \Sigma \\ \iff \mathbb{E}(Y(X - \mu)^T) &= \beta_1^T \Sigma, \end{aligned}$$

with Σ being the covariance matrix of the predictors. Setting $\sigma_{YX} = \mathbb{E}(Y(X - \mu)^T)$ and using the symmetry of Σ , we yield $\beta_0^* = \mathbb{E}(Y)$ and $\beta_1^* = \Sigma^{-1}\sigma_{YX}$.

Plugging in of these solutions into $m(x)$ and some rearranging of the terms in $\mathbb{E}(Y)$ and σ_{YX} yields the term that was to show. ■

Replacing the Euclidean distance above with a general metric yields the general Fréchet regression model.

Post-submission addition: from the Euclidean case to general metric spaces. *Not part of the thesis submitted in October 2023.*

The lemma above characterises the Euclidean conditional mean as the minimiser of a *weighted* least-squares criterion whose weights $s(z, x)$ depend on the predictor only through the first two moments of X . The global Fréchet regression model of Petersen and Müller (2019) is obtained by keeping these weights and replacing the Euclidean distance d_E by a general metric d .

Definition 5.2 (Global Fréchet regression)

Let (Ω, d) be a metric space, let $X \in \mathbb{R}^r$ be a predictor with mean μ and non-singular covariance matrix Σ , and define

$$s(z, x) = 1 + (z - \mu)^\top \Sigma^{-1} (x - \mu).$$

The *global Fréchet regression function* is

$$m_\oplus(x) = \arg \min_{\omega \in \Omega} \mathbb{E} \left[s(X, x) d^2(Y, \omega) \right]. \quad (20)$$

Given a sample $(X_i, Y_i)_{i=1}^n$, write $\bar{X}$ for the sample mean and $\hat{\Sigma}$ for the sample covariance of the X_i , and set

$$s_{in}(x) = 1 + (X_i - \bar{X})^\top \hat{\Sigma}^{-1} (x - \bar{X}).$$

The corresponding estimator is

$$\hat{m}_\oplus(x) = \arg \min_{\omega \in \Omega} \frac{1}{n} \sum_{i=1}^n s_{in}(x) d^2(Y_i, \omega). \quad (21)$$

Setting $\Omega = \mathcal{G}$ and $d = d_W$ specialises this to our context. Because $d_W = d_Q$ by Lemma 2.10, the criterion (21) becomes a weighted average of squared $\mathcal{L}^2$ distances between quantile functions,

$$\hat{m}_\oplus(x) = \arg \min_Q \frac{1}{n} \sum_{i=1}^n s_{in}(x) \int_0^1 \left(Q_i(u) - Q(u) \right)^2 du, \quad (22)$$

whose *unconstrained* minimiser is the weighted cross-sectional mean of the sample quantile functions,

$$\tilde{Q}(x)(\cdot) = \frac{1}{n} \sum_{i=1}^n s_{in}(x) Q_i(\cdot).$$

When $\tilde{Q}(x)$ is non-decreasing it is already a valid quantile function and hence the solution; otherwise $\hat{m}_{\oplus}(x)$ is its $\mathcal{L}^2$ -projection onto the cone of non-decreasing functions, which is precisely the quadratic program in Algorithm 3.

Next, we choose the distance to be d_W and the space of Y to be $\mathcal{G}$ as in Definition 2.8.

5.2. Computational Details

We used this algorithm provided in the supplementary material of Petersen, Liu, et al. (2021). It first checks whether our estimator is a valid qf, otherwise it finds the closest qf to the estimator via quadratic programming. Here,

$$s_{in}(p) = 1 + (X_i - \bar{X})^T \hat{\Sigma}^{-1}(p - \bar{X})$$

denotes the empirical version of the weight function from Lemma 5.1, with $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T$ (cf. Petersen and Müller, 2019).

Algorithm 3 Estimating $\hat{Q}(p)$

Require: Predictor vector $p \in \mathbb{R}^r$; quantile functions Q_i , and grid $0 = u_1 < \dots < u_m = 1$

Ensure: Estimates $\hat{Q}(p; u_l); l = 1, \dots, m$

- 1: **for** $l = 1, \dots, m$ **do**
- 2: Compute $Q_l = \frac{1}{n} \sum_{i=1}^n s_{in}(p) Q_i(u_l)$
- 3: **end for**
- 4: **if** $Q_{l+1} \geq Q_l$ for all $l \in \{1, \dots, m-1\}$ **then**
- 5: Set $\hat{Q}(p; u_l) = Q_l$
- 6: **else**
- 7: Compute $b^* = \min_{b \in \mathbb{R}^m} \frac{1}{2} b^\top A b - Q^\top A b$ subject to $b_1 \leq \dots \leq b_m$
- 8: Set $\hat{Q}(p; u_l) = b_l^*$
- 9: **end if**

Results for this method are shown in Section 7.

6. A Common Transformation Framework

Post-submission section. *Not part of the thesis submitted in October 2023.*

This thesis has so far presented two routes to density regression in two different languages: Section 4 works with the log quantile density (LQD) transformation in the Hilbert space $\mathcal{L}^2([0, 1])$, whereas Section 5 works intrinsically in the Wasserstein space $(\mathcal{G}, d_W)$. In this section we place both methods in a single template, which makes their similarities precise and gives a geometric explanation for the gap between them observed in the simulation study of Section 7.

6.1. A common template

Both methods share the structure

$$\mathcal{G} \xrightarrow{\Phi} \mathcal{H} \xrightarrow{\text{linear regression}} \mathcal{H} \xrightarrow{\Phi^{-1}} \mathcal{G}, \quad (23)$$

that is, a *representation map* (or chart) Φ sends a density to an element of a Hilbert space $\mathcal{H}$, a linear regression is carried out there, and the fit is mapped back to $\mathcal{G}$ for interpretation. The two methods differ only in the choice of Φ .

LQD method. Here $\Phi = \psi$ is the LQD transformation (9) and $\mathcal{H} = \mathcal{L}^2([0, 1])$. The image is unconstrained, so the regression reduces to coefficientwise ordinary least squares. The price is twofold: ψ is a nonlinear map, and it is *blind to location*, since a density f and any translate $f(\cdot - c)$ share the same quantile density (cf. Section 4.1.2). The location must therefore be carried as a separate scalar $c = Q(u^*)$ and reconstructed on inversion.

Fréchet method. Here $\Phi = Q$ is the quantile representation and again $\mathcal{H} = \mathcal{L}^2([0, 1])$, but the image is restricted to the closed convex cone $\mathcal{Q}$ of non-decreasing functions. By Lemma 2.10 the map $\Phi : (\mathcal{G}, d_W) \rightarrow (\mathcal{Q}, \|\cdot\|_2)$ is an *isometry* (cf. Panaretos and Zemel, 2020, Chapter 2). As shown in Definition 5.2, global Fréchet regression is then ordinary linear regression on the quantile functions Q_i inside this cone, followed by the $\mathcal{L}^2$ -projection onto $\mathcal{Q}$ of Algorithm 3. Because the location is encoded in Q itself, no separate anchor is needed.

Table 1 summarises the correspondence.

	LQD method	Fréchet method
Representation Φ	$\psi = \log q$	Q (quantile function)
Image space $\mathcal{H}$	$\mathcal{L}^2([0, 1])$	cone $\mathcal{Q} \subset \mathcal{L}^2([0, 1])$
Constraints in $\mathcal{H}$	none	monotonicity (convex cone)
Location information	lost; anchor $c = Q(u^*)$	retained in Q
Regression step	coefficientwise OLS	weighted mean of Q_i
Return to $\mathcal{G}$	ψ^{-1} and anchor	projection onto $\mathcal{Q}$
Relation to d_W	nonlinear distortion	isometry

 Table 1: The LQD and Fréchet methods as two instances of the template (23), differing only in the representation map Φ .

6.2. Why the geometry favours the Fréchet method

The two charts are not interchangeable, because the error criterion is itself defined in one of them. The integrated squared error reported in Section 7 is measured in d_W , and by Lemma 2.10 this equals the squared $\mathcal{L}^2$ distance between quantile functions. The Fréchet method therefore regresses in *exactly* the chart in which the loss is the flat $\mathcal{L}^2$ metric. Since $\Phi = Q$ is an isometry, a linear model in Q -space is consistent with the Wasserstein geometry: minimising in-chart squared error coincides with minimising d_W risk. The only departure from a plain linear fit is the projection onto the cone $\mathcal{Q}$; but projection onto a closed convex set in a Hilbert space is non-expansive, and the true conditional quantile function $m_{\oplus}(x)$ lies in $\mathcal{Q}$, so

$$\|\Pi_{\mathcal{Q}}(\tilde{Q}(x)) - m_{\oplus}(x)\|_2 \leq \|\tilde{Q}(x) - m_{\oplus}(x)\|_2, \quad (24)$$

where $\Pi_{\mathcal{Q}}$ denotes the projection onto $\mathcal{Q}$. The monotonicity correction can thus only reduce the d_W error: it is, in this sense, “free”.

For the LQD method the situation is different. The chart $\Phi = \psi$ is a nonlinear, non-isometric distortion of $(\mathcal{G}, d_W)$. A linear model that is optimal in ψ -space — that is, in $\mathcal{L}^2$ error on the log quantile densities — is in general *not* optimal for d_W risk, because ψ does not preserve the metric. In addition, the location component discarded by ψ must be reinstated from the separately estimated anchors $c(p)$ (here by linear interpolation), and any error in this reconstruction enters the d_W risk directly and does not vanish as the fit of the log quantile densities improves.

This gives a geometric, rather than merely computational, explanation for the persistent gap in Table 2. Even when the densities are observed directly, the LQD risk is dominated by representation and location error, whereas the Fréchet risk reflects only estimation error in the chart aligned with the loss. The density-estimation step adds a comparable amount of noise to both methods but does not change this ordering. In short, the Fréchet method performs better here not by an accident of tuning, but because it linearises the problem in the same geometry in which performance is measured.

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In this section, we show how the LQD method and Fréchet method work in practice and compare their performance in a simulation study. First, we will describe the DGP that will produce the data for our illustrations and describe the density estimation step. Before turning to the results of the simulation study, we will use a sample dataset generated by this DGP to illustrate the behavior of the two methods.

7.1. Data Generating Process

The DGP used here is (almost) the same as Petersen and Müller (2019) used in their first simulation scenario. We generate a sample of qfs that is positively linearly dependent on our predictor p . The basis is a normal qf whose location and scale both rise linearly in p . The qf-valued stochastic process is given as follows:

$$m_{\oplus}(x)(\cdot) = \mathbb{E}[Y(\cdot)|X = x] = \mu_0 + \beta x + (\sigma_0 + \gamma x)\Phi^{-1}(\cdot),$$

where Φ is the cdf of the standard normal distribution. Compared to the version of the authors, we slightly exaggerated the parameters for more pronounced visualization. In our case $\beta = 4$ and $\gamma = 2$, while having μ_0 and σ_0 both still equal to 0 and 3, respectively. The sample is generated as follows. First we sample the predictor $p \sim U([-1, 1])$. Then we draw the corresponding location and scale parameters as follows

$$\mu|X \sim \mathcal{N}(\mu_0 + \beta X, v_1)$$

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$$\sigma|X \sim \text{Gam}\left(\frac{(\sigma_0 + \gamma X)^2}{v_2}, \frac{v_2}{\sigma_0 + \gamma X}\right).$$

Sampling the scale from a Gamma distribution ensures drawing positive parameters, as is required. The values for v_1 and v_2 are 0.25 and 2, respectively. Drawing μ and σ like this adds some noise to the observations: μ oscillates with standard deviation v_1 , and σ with variance v_2 , around the respective true values.

More details can be found in the paper. We now briefly mention the density estimation step.

7.2. Density Estimation

Since we will have a scenario where we have to estimate the densities f_i from a finite sample $x_{i1}, \dots, x_{im}$ of observed points for each of the f_i . The densities were estimated using a kernel density estimator with standard normal kernel. This gives additional uncertainty and we will see in the simulation results if the estimates by the different methods respond differently to this uncertainty though we do not expect them to.

7.3. Illustration of Methods in Example

Before turning to the simulation study, we briefly illustrate the behavior of the two methods for a given sample generated with the DGP described above.

7.3.1. LQD method

Here, we show the behavior of the LQD method in estimating the true stochastic process. Figure 2 shows the estimated $\psi(\hat{\beta})$ values. $\psi(\hat{\beta}_1)$ is estimated to be a constant greater than one, meaning that $\hat{\beta}_1$ is a positive constant after back transformation, reflecting the positive linear effect that p has on the responses. $\psi(\hat{\beta}_0)$ reflects the mean LQD, or $\hat{\beta}_0$ reflects the mean pdf. Calculating fitted values via $\psi(\hat{f}) = \psi(\hat{\beta}_0) + \psi(\hat{\beta}_1)p$ means that we are adding an increasing constant for increasing p to a (log) qdf, and this constant makes the qdf values uniformly larger everywhere, thus making them relatively more even. A flatter qdf means more variance so we should see a gradual increase in variance in the plots of the estimated pdfs via LQD method. The location, as mentioned in Section 4.1.2, has to be estimated separately and is not contained in the information of qdfs, so also

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not in the estimated LQDs or regression coefficients. To show that $\psi(\hat{\beta}_0)$ and the (cross-sectional) mean LQD function are the same, we printed Figure 3. Next, we will turn to the estimates of the densities obtained via LQD method and compare them with the densities produced by the true stochastic process. Figure 4 plots the estimated densities against their true counterparts. We see that the general trend to higher variance is well reflected in the estimated curves but the estimation of the support location is taking its toll on accuracy.

Next, we will briefly show the performance of the Fréchet mean.

7.3.2. Fréchet method

Figure 5 shows two example qfs, generated from the above mentioned process with the predictor values $p = -1$ and $p = 1$. One can see that both location and scale are much larger for the curve with the higher predictor value.

Figure 6 plots the corresponding pdfs of some of the generated qfs.

Figures 7 and 8 plot the Fréchet estimates of the pdfs and qfs respectively. In the first figure, the estimated pdfs for different predictor values are drawn on top of each other, and in the second, a 3d plot was used to visualize the change of shape of the qfs. One can see a smooth increase in location and variance in both figures with an increase in the predictor value.

Figure 9 shows a more continuous estimate in the Fréchet regression. Clearly the location shifts steadily to the right, as x increases, and the variance increases as well.

7.4. Results

In this section, we briefly visualize and discuss the results from the simulation study.

Figure 10 shows the distribution of ISE values for the Fréchet method, in the scenario where the densities are directly observed. One can clearly see that, with increasing sample size, they become lower and have a lower variance, with less outliers outside the whiskers. This is as expected, since increasing the sample size means more information to use for estimation, so the predictions should be closer to their mean (the true value), and the variance of the predictions around their mean should get smaller.

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Figure 11 shows the distribution of ISE values for the Fréchet method, in the scenario with a previously included density estimation step. Here, the ISE values are much higher, and increasing sample sizes yields only minor improvement. This hints at a problem in the density estimation method. Nonetheless, the values are better than the ISE values for the transformation method in the same scenario.

Finally, Figure 12 shows the distribution of ISE values for the functional regression method, in both scenarios and for different sample sizes. One can see that the added uncertainty by the density estimation step lowers the convergence rate, and that in both scenarios the ISE gets stuck on a high level as increasing the sample size from 100 to 200 makes negligible difference. This is probably because of additional tuning that has to be done in the density estimation step (since this scenario makes the Fréchet method also significantly worse), such as a better method of bandwidth choice. The big difference between the ISE boxplots for the Fréchet method and the transformation method even in the case of directly observed densities must be explained by the aforementioned (see Section 4.1.2) need to estimate the location of the support in the transformation method since this information is lost when transforming the pdf to the qdf. An area of improvement could be to use more sophisticated estimation methods than linear interpolation to compute the left bounds of the supports.

ISE	$n = 50$	$n = 100$	$n = 200$
Fréchet	0.18	0.08	0.04
Fréchet, D.-Est.	32.31	27.98	27.38
LQD	34.49	31.19	31.88
LQD, D.-Est.	47.08	43.39	43.43

Table 2: Mean ISE Values for different methods and scenarios. "D.-Est." indicates a scenario with previous density estimation procedure.

Table 2 shows the mean ISE values from the different simulation settings, for both methods. They reflect the same findings discussed in the boxplots above. Moreover, the quality of the density estimates deteriorates with sample size, suggesting to vary the bandwidth choice with sample size, as Petersen and Müller (2019) did in their comparison of the Fréchet estimator with the Nadaraya-Watson estimator. Due to time constraints and the computational requirements, we did not conduct such hyperparameter tuning but chose a rule-of-thumb bandwidth (Silverman, 1986, Chapter 3.4.1; Li and Racine, 2007, Chapter 2.2.1).

7.5. Additional Scenarios

The DGP examined so far favors the Fréchet method in two distinct ways. First, the true regression function is linear in the space of qfs, which is precisely the model that global Fréchet regression postulates, so the Fréchet method is correctly specified while the functional linear model in LQD space is not. Second, the supports of the sampled densities differ, so the LQD method has to estimate the location of the support, which, as discussed above, dominates its error. To disentangle these two effects, we consider three additional scenarios, all in the setting of directly observed densities and with $p \sim U([-1, 1])$ as before. Each scenario was run with 200 Monte Carlo repetitions; the implementation can be found in the file *simulation_new_scenarios.py* in the repository.

In the second scenario, we remove the support estimation while keeping the truth linear in the space of qfs. We perturb the uniform distribution on $[0, 1]$ with random transport maps (cf. Panaretos and Zemel, 2016): conditional on the predictor, the response has the qf

$$Q(u) = u + a \frac{\sin(2\pi u)}{2\pi}, \quad a = \alpha p + \varepsilon, \quad \varepsilon \sim U(-\nu, \nu),$$

with $\alpha = 0.5$ and $\nu = 0.25$, so that all sampled maps are strictly increasing and all densities share the common support $[0, 1]$. Since $\mathbb{E}[a | p] = \alpha p$, the true conditional Fréchet mean is again linear in the space of qfs and the Fréchet method remains correctly specified. The LQD method is misspecified — the LQD of these densities, $\log(1 + a \cos(2\pi u))$, is not linear in a — but no longer needs to estimate the location of the support.

In the third scenario, we mirror the situation such that the functional regression model after LQD transformation is the correctly specified one. We generate the LQDs directly from the functional linear model

$$\psi_i(u) = p_i \cos(\pi u) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.1^2),$$

and obtain the qfs as $Q_i(u) = \int_0^u \exp\{\psi_i(s)\} ds$, so that all supports start at zero. The true regression function is exactly linear in LQD space, while the conditional qfs are nonlinear in the predictor, so now the Fréchet method is misspecified.

The fourth scenario uses the location-scale process of the first scenario but makes both parameters nonlinear in the predictor: the conditional location is $4 \sin(\pi p)$ and the conditional scale is $3 + 2p^2$, with the same noise structure as before. Since both global Fréchet

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regression and the functional linear model in LQD space are linear in the predictor, both methods are misspecified in this scenario.

Scenario	Method	$n = 50$	$n = 100$	$n = 200$
2: Common support ($\times 10^{-3}$)	Fréchet	0.026	0.012	0.005
	LQD	0.360	0.327	0.304
3: Linear in LQD space ($\times 10^{-3}$)	Fréchet	4.315	4.019	3.851
	LQD	0.330	0.210	0.084
4: Nonlinear	Fréchet	7.485	7.205	7.099
	LQD	28.708	30.793	30.972

Table 3: Mean ISE values in the additional scenarios, based on 200 Monte Carlo repetitions with directly observed densities. The values for scenarios 2 and 3 are scaled by 10^3 .

Table 3 shows the mean ISE values for the three additional scenarios. The results confirm the interpretation of the first scenario. In the common support scenario, the Fréchet method, being correctly specified, is still more accurate, and its error vanishes at a fast rate with growing sample size. The error of the LQD method, in contrast, settles at a low but constant level of about 3×10^{-4} — its misspecification bias. The instructive comparison, however, is with the first scenario: there, the mean ISE of the LQD method was larger than that of the Fréchet method by a factor of almost 800 (at $n = 200$), while here the factor is around 60 on an overall much smaller scale. Once the common support removes the need to estimate the support location, the LQD method becomes competitive, which confirms that almost all of its error in the first scenario stems from the support estimation rather than from the regression step itself.

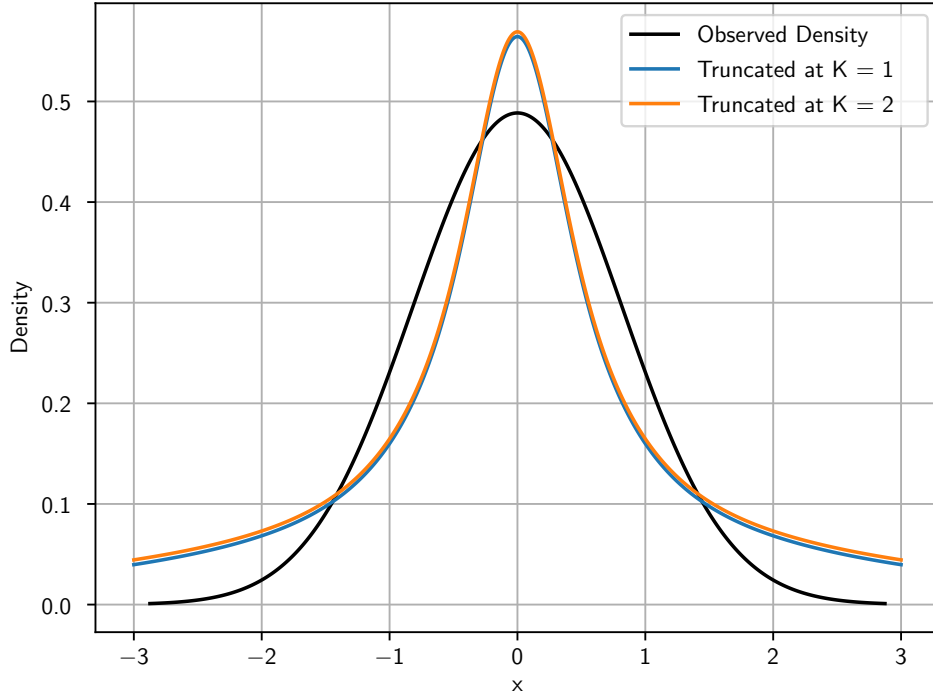
In the third scenario the roles are exactly reversed. Now the error of the LQD method decreases steadily with the sample size, as its regression model in the transformed space is correctly specified, while the Fréchet method settles at a misspecification bias of about 4×10^{-3} , roughly 45 times the LQD error at $n = 200$. Notably, both methods remain accurate in absolute terms: the misspecified Fréchet estimates are still proper distributions close to the truth, whereas the misspecified LQD method in the second scenario at least required the common support to achieve the same.

In the fourth scenario, where neither model is correct, both errors are large and do not improve with the sample size, as they are dominated by bias. The Fréchet method degrades much more gracefully: its mean ISE of about 7 is less than a quarter of that of the LQD method, whose error remains at the same level as in the first scenario since the support estimation again dominates. A natural remedy for both methods in such

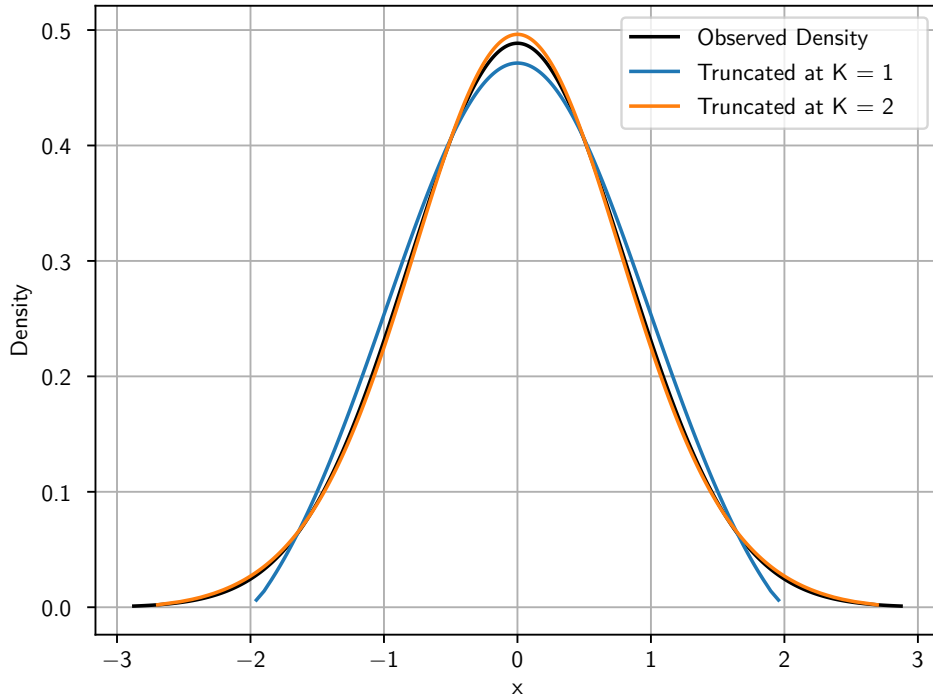
7. *Application*

settings would be a local regression variant, such as the local Fréchet regression or the Nadaraya–Watson-type estimator considered by Petersen and Müller (2019).

7. Application



(a) Naive method.



(b) LQD method.

Figure 1: Truncated Karhunen-Loève representations of a sample density. Figure 1a shows the representations with the naive method, Figure 1b shows it for the LQD method. The blue and orange curves result from using one and two principal components, respectively. The black curve is the sample density that is to be represented. We added a small perturbation to the orange curve in the naive method to better differentiate it from the blue curve.

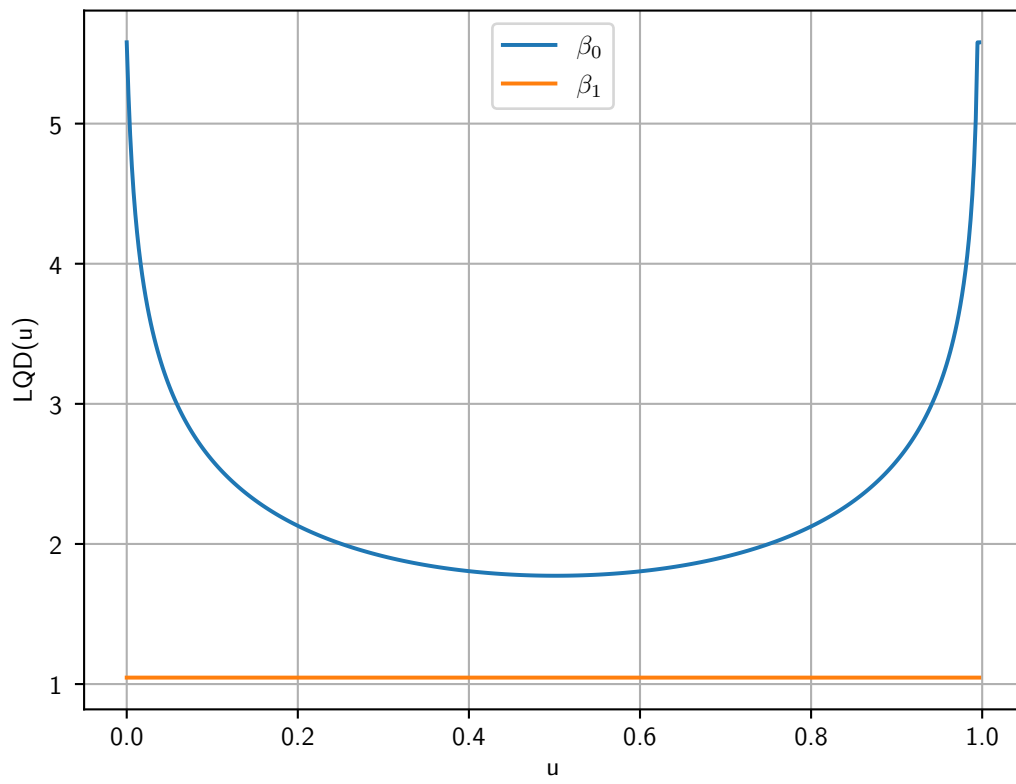


Figure 2: Plot of estimated betas from functional regression with LQD method.

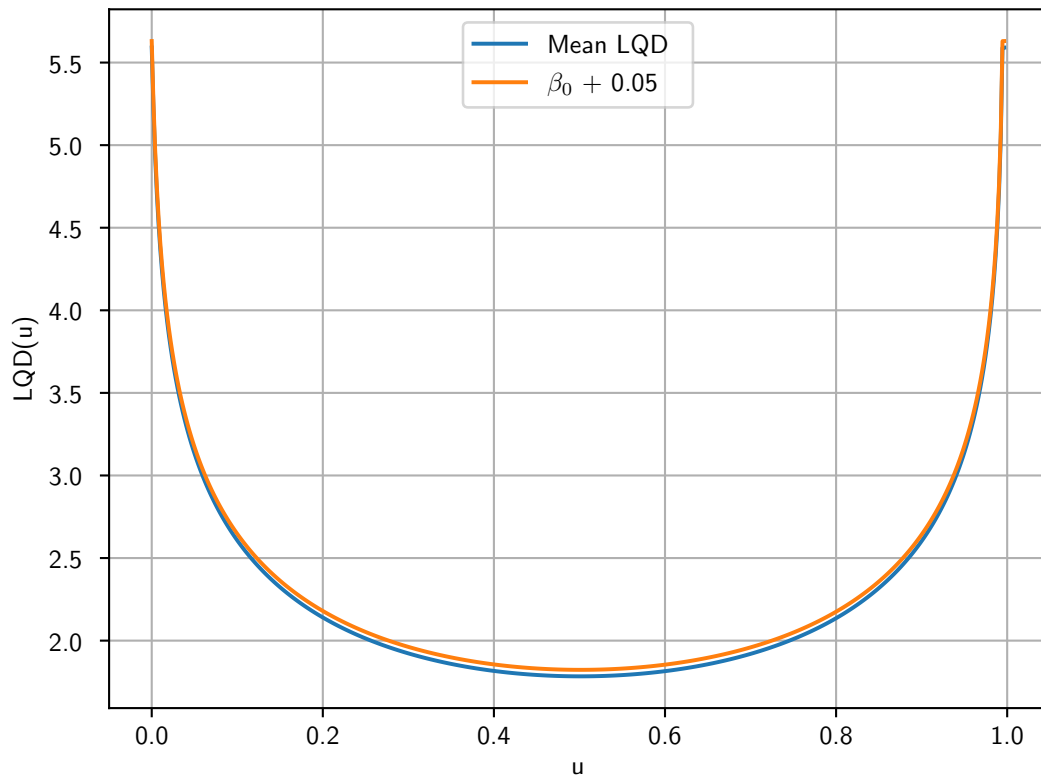


Figure 3: Plot that shows the equality of the mean LQD function and β_0 . Some perturbation was added so one can distinguish the curves.

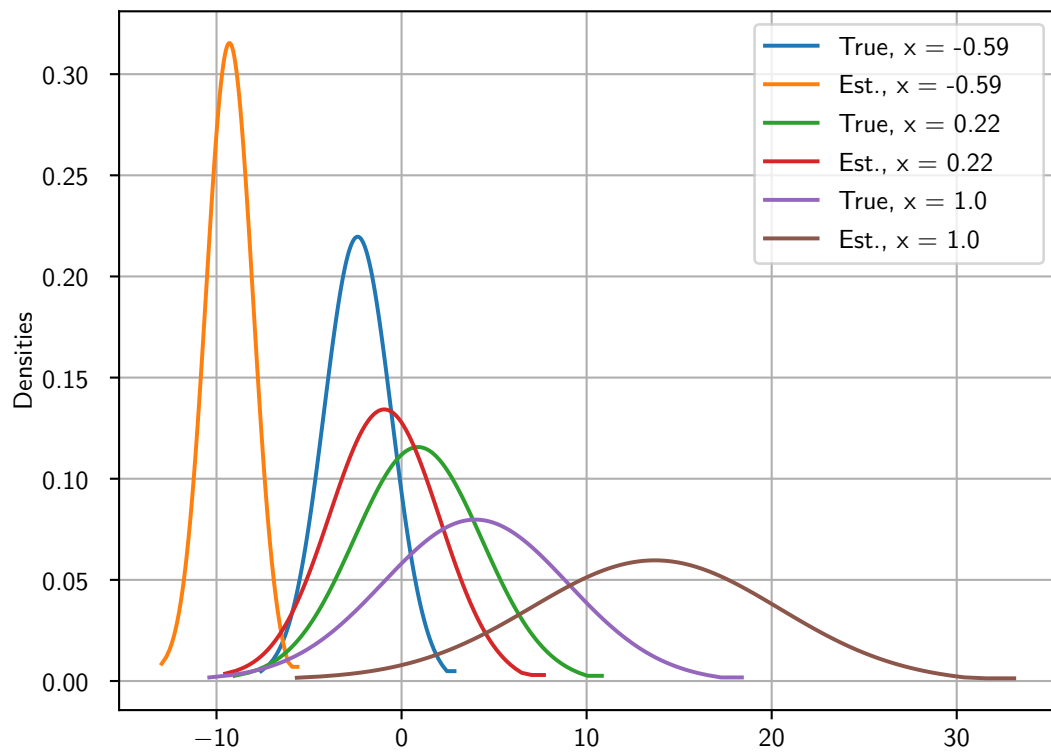


Figure 4: Estimated densities from functional regression with LQD method plotted against the true densities for some selected predictor values.

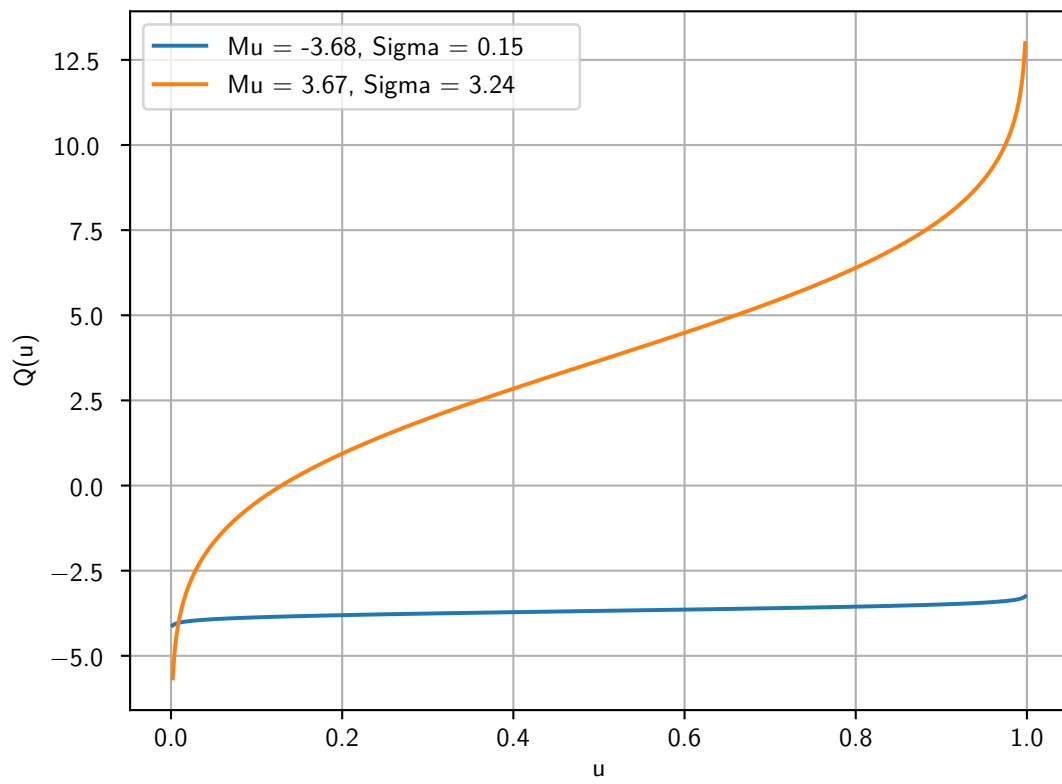


Figure 5: Plot of two sample qfs, one with predictor value $p = -1$, and the other for $p = 1$.

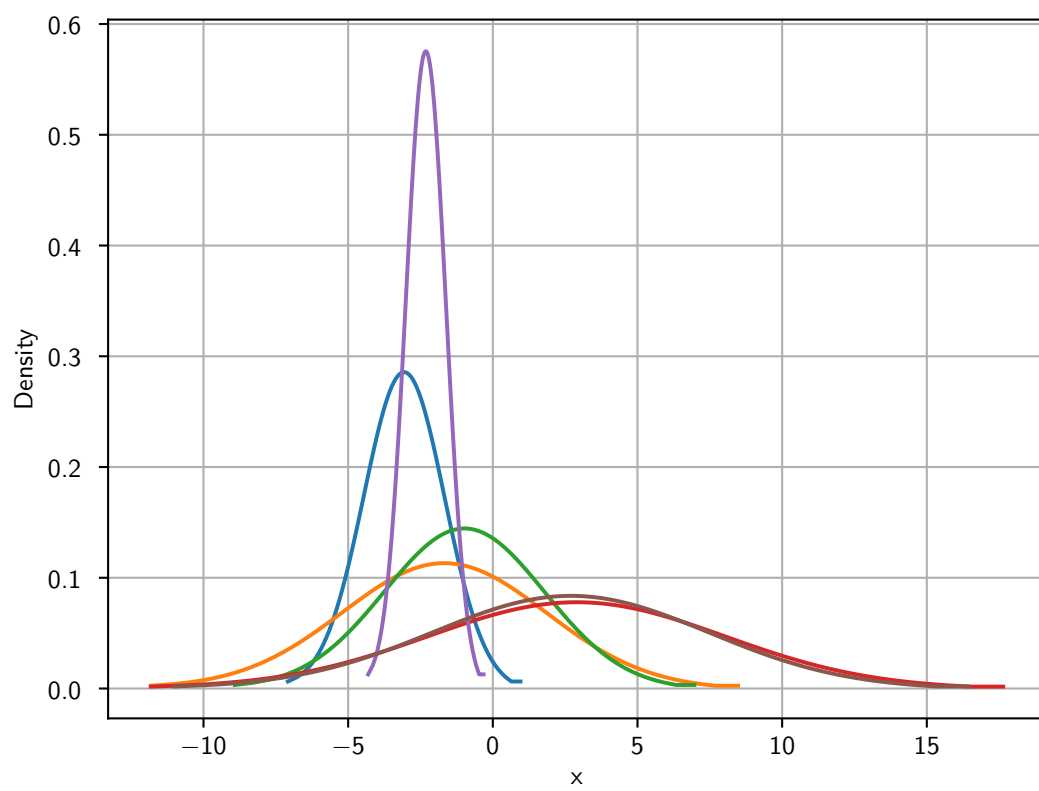


Figure 6: Plot of several sample pdfs.

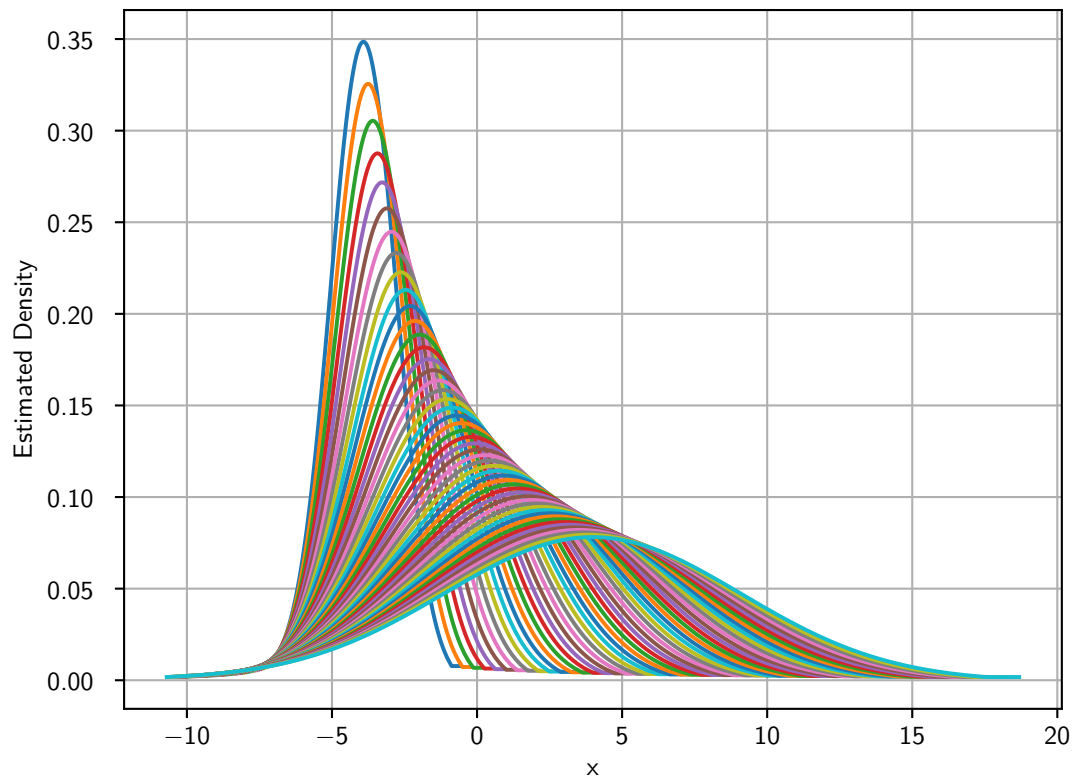


Figure 7: Plot of estimated densities via Fréchet regression method. The densities from left to right are corresponding to increasing predictor values p on an equidistant grid on $[-1, 1]$ with 50 grid points.

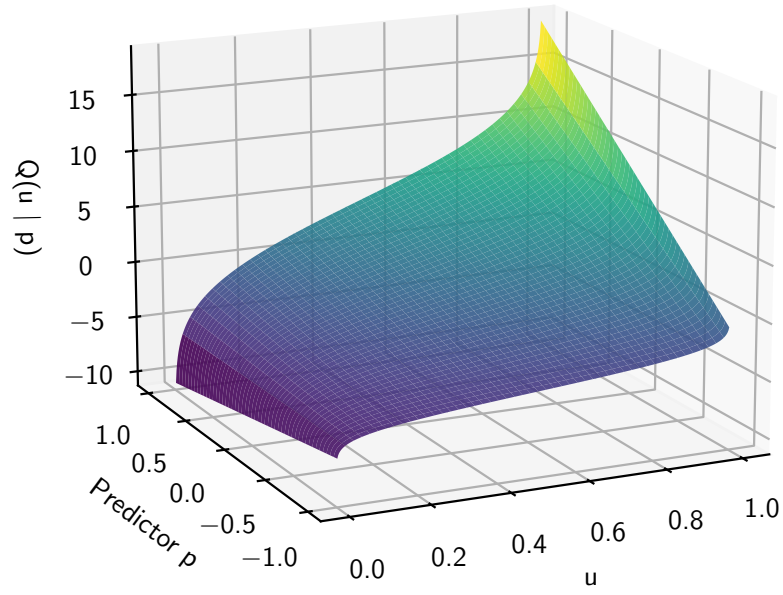


Figure 8: Plot of estimated qfs via Fréchet regression method.

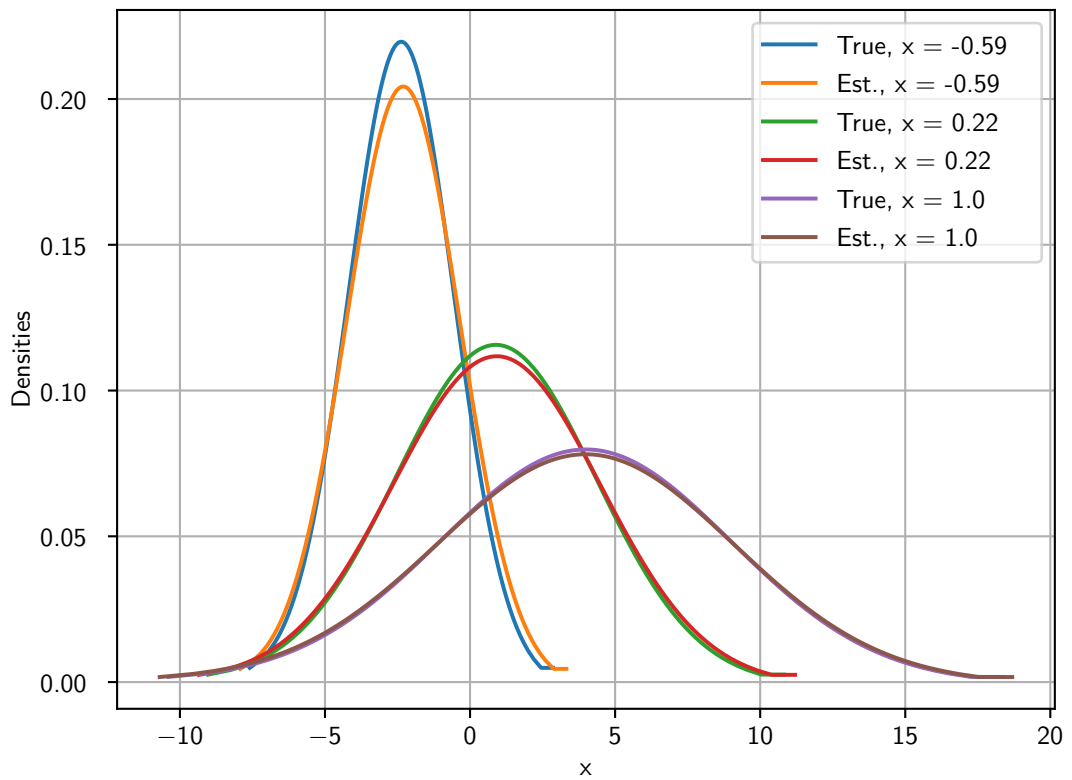


Figure 9: Estimated densities from Fréchet regression plotted against the true densities for some selected predictor values.

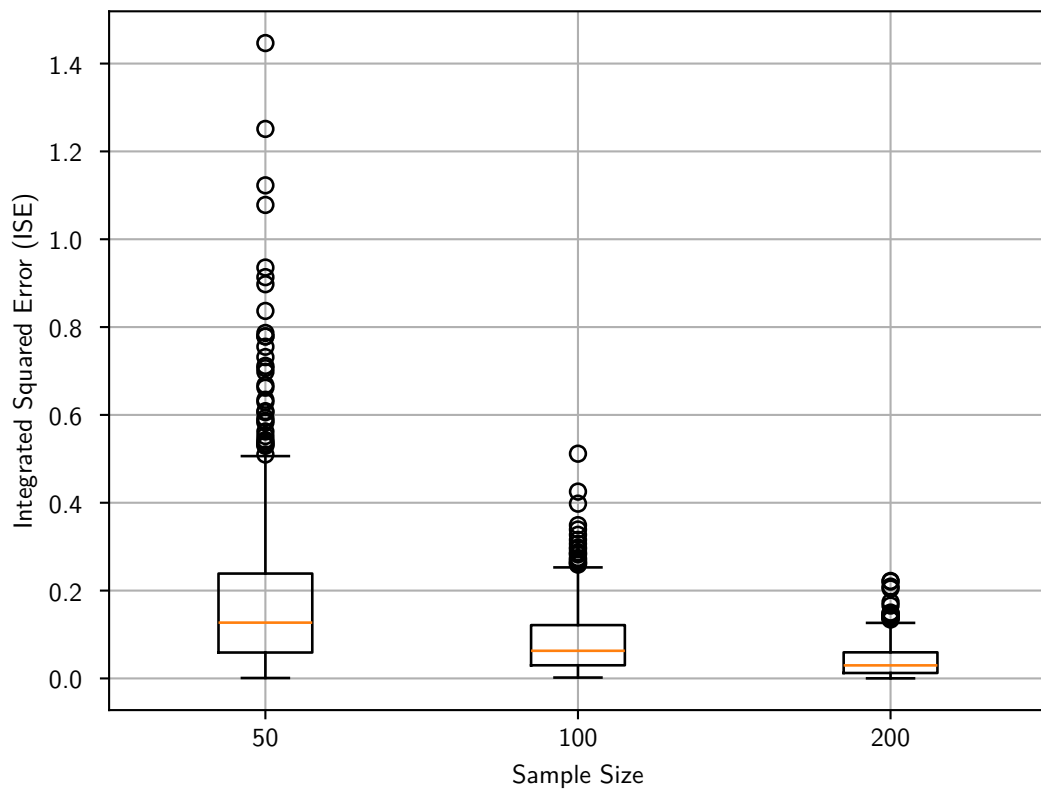


Figure 10: Boxplot showing distribution of ISE from Fréchet estimates in scenario with directly observed densities.

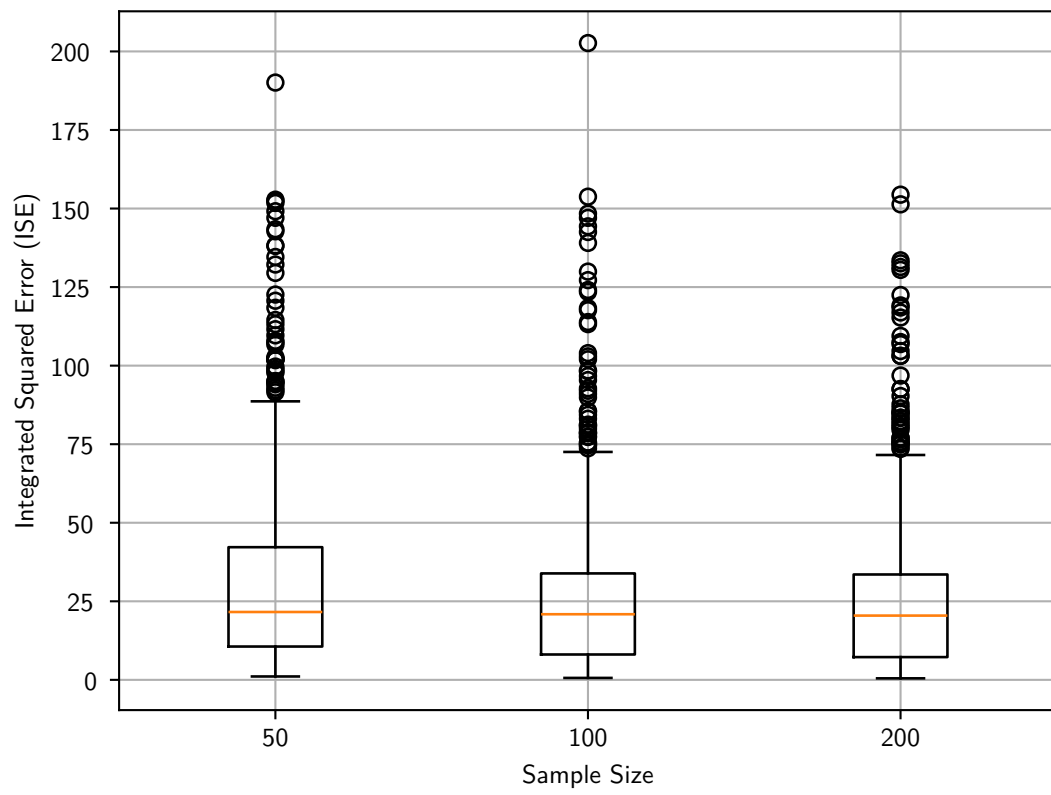


Figure 11: Boxplot showing distribution of ISE from Fréchet estimates in scenario with additional density estimation step to obtain densities.

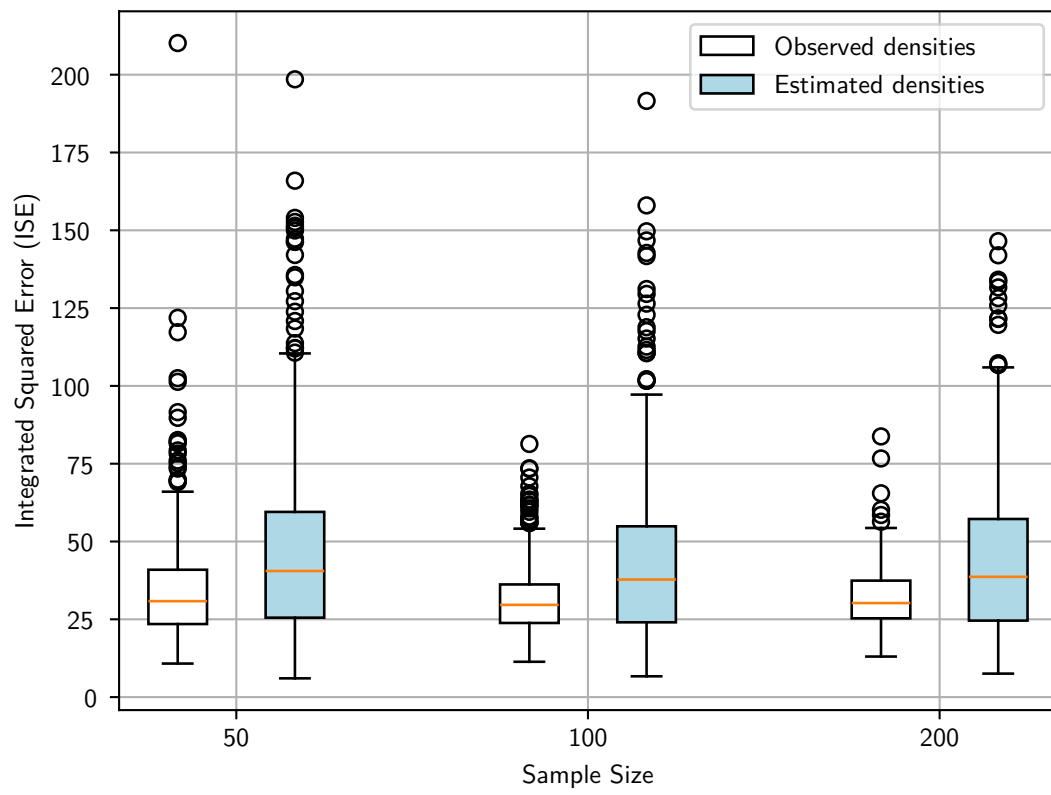


Figure 12: Boxplot showing distribution of ISE from functional regression with LQD method in both scenarios.

8. Conclusion

In this thesis, we compared two approaches to regression with density responses: the functional regression model after transformation to the space of log quantile densities (Petersen and Müller, 2016; Kokoszka, Miao, et al., 2019), and global Fréchet regression in the Wasserstein space (Petersen and Müller, 2019).

The simulation study makes the practical trade-off between the two methods concrete. When the true regression function is linear in the space of quantile functions and the densities have differing supports (Section 7.1), Fréchet regression is both correctly specified and unburdened by nuisance parameters: its mean ISE falls from 0.18 to 0.04 as the sample size grows from 50 to 200, while the LQD method remains stuck above 31. The additional scenarios in Section 7.5 disentangle why. With a common support, the very same comparison shrinks from a factor of several hundred to well within an order of magnitude, showing that almost all of the LQD method’s error is caused by having to re-estimate the support location that the transformation discards — not by the regression in the transformed space itself. Conversely, when the truth is linear in LQD space, the LQD method wins, but the misspecified Fréchet estimator still settles at a small bias of about 4×10^{-3} ; when neither model holds, the Fréchet method degrades far more gracefully (mean ISE of about 7 against about 31). Finally, an estimation step for the densities themselves dominates the error of both methods and erodes their convergence, underlining the importance of the bandwidth choice discussed in Section 7.4.

Beyond the numbers, the two methods differ markedly in their ease of application. Fréchet regression adapts to the geometry of the Wasserstein space directly, needs no transformation and back-transformation, produces valid distributions by construction, and required no tuning in our implementation beyond the monotonicity projection of Algorithm 3. The transformation method is more delicate: it requires the choice of an anchor point and an estimator for it, inherits the numerical fragility of quantile densities near the boundaries (Appendix A), and its results are only as good as the linearity assumption in the transformed space. Its strength is the converse of that assumption: when a functional linear model in LQD space is plausible — and the supports are common or anchored — it attains the better rate, and it gives access to the full fda toolbox in the transformed space.

Several extensions suggest themselves. Local Fréchet regression or Nadaraya–Watson-type smoothing would address the bias that both global methods exhibit under the nonlinear

8. Conclusion

DGP of Section 7.5. Richer families of transport maps (Panaretos and Zemel, 2016) could generate more demanding shape variation than our sine-perturbation scenario. For inference rather than point estimation, Petersen, Liu, et al. (2021) provide confidence bands for Fréchet regression in the Wasserstein context. And on the practical side, a sample-size-dependent bandwidth choice in the density estimation step, together with a more sophisticated estimator of the support location than linear interpolation, would give the comparison in the estimated-density scenarios sharper contours.

In summary, when densities are the response, global Fréchet regression is the more robust default: it is simpler to apply, never leaves the density space, and its losses under misspecification are modest, whereas the transformation method's losses under support uncertainty are not.

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A. Numerics

For densities with smaller sigmas, we get very low density values for large parts of the support, leading to numerical artifacts when calculating the quantile densities (infs near the right bound, astronomical values near the left bound) since the qfs corresponding to those low variance pdfs have a very steep slope near the boundaries. These artifacts break further calculations and make it impossible to get the original densities from those broken qds again. We chose to calculate the "effective" support of the densities, if they have very small sigmas, defined by the density values being larger than some small $\delta > 0$ on this support (see Appendix B.2). This makes the transformation method from Petersen and Müller (2016) inadequate (which assumes a common support), hence we used the modified transformation from Kokoszka, Miao, et al. (2019). This is described in more detail in Section 4.1.2.

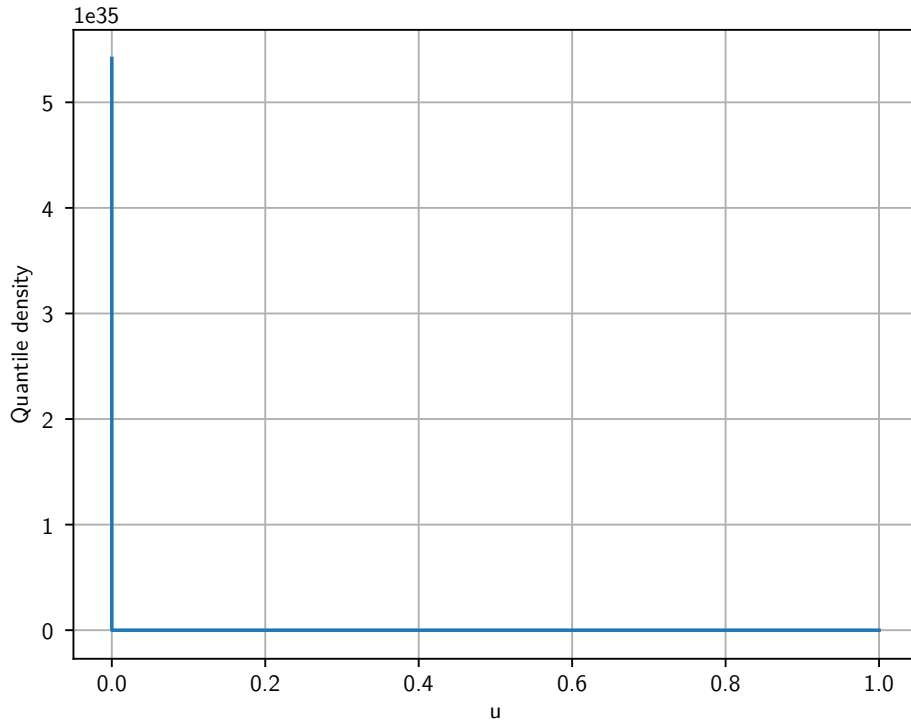


Figure 13: A numerically broken qdf from a low-variance distribution. The values near the right boundary became too large to store for the computer.

Figure 13 shows one of the broken qdfs, and Figure 14 shows the fixed version of it by truncating the support of the pdf:

In their own R package (*fdadensity*), the authors have a "RegularizeByAlpha" function, which tries to raise the minimal value of a density to a given level alpha. This also gets rid

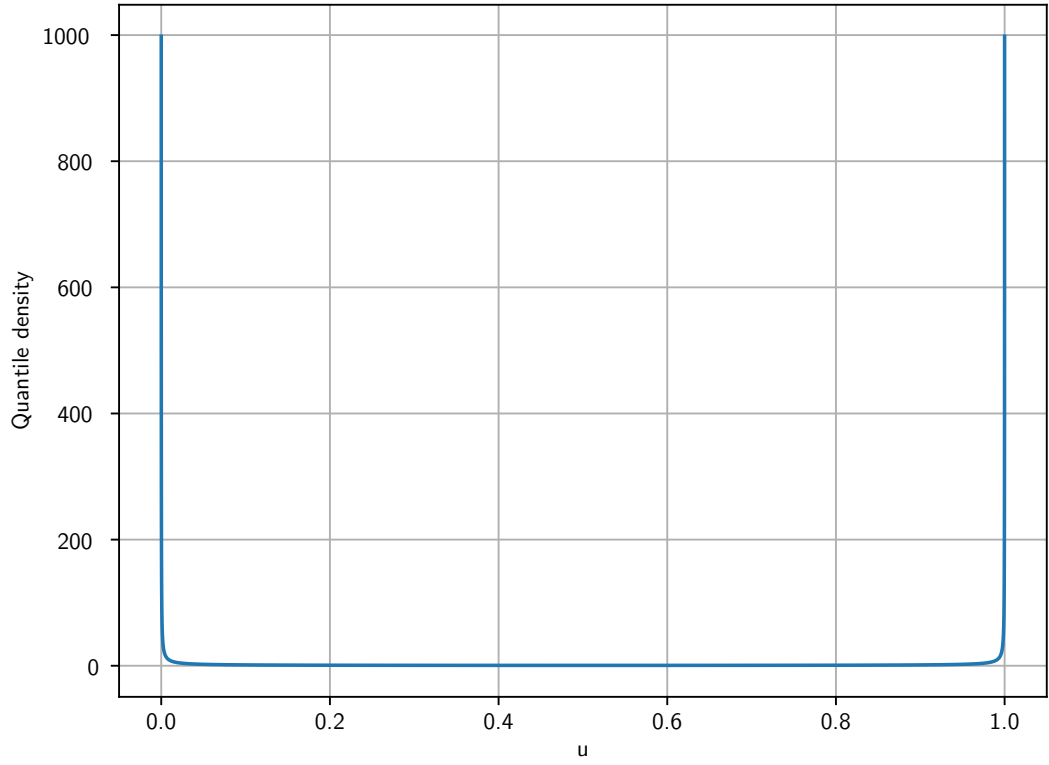


Figure 14: Fixed qdf by truncating the pdf support to where the pdf is greater than $\varepsilon = 10^{-3}$.

of the numerical artifacts and does not change the support of the function, which makes the methods from their paper still adequate. At the end of the calculations, they revert this change with "DeRegularizeByAlpha".

B. Code

In this chapter, some code snippets are provided for the most important functions we used.

Central in our code is the class *Function* that we defined. It can hold an analytical function, with attributes *x* (containing the domain values), *y* (containing the function values), and *grid_size* (the number of discretization points at which the function is evaluated), among others. It has many different methods, like *integrate()*, *differentiate()*, *invert()* etc., that are used throughout the code and yield new Function class objects. It can be inspected in the "function_class.py" file in the GitHub repository (https://github.com/mirkimirk/thesis_frechet_fda) and would be too large to print here.

B.1. Python Code for pdf to qdf Conversion

The following code shows how we converted pdfs to qdfs and vice versa.

The option *center_on_zero* in the *qdf_to_pdf* function artificially centers the resulting pdf around zero. It is not used in any of the plots in this thesis.

```

1      def pdf_to_qdf(pdf: Function, save_support_start: bool = False) ->
      Function:
2          """Directly convert a pdf to a qdf using inverse function rule
          on qf."""
3          quantile_func = pdf.integrate().invert()
4          if save_support_start:
5              return (1 / pdf.compose(quantile_func), quantile_func.y[0])
6          else:
7              return 1 / pdf.compose(quantile_func)
8
9      def qdf_to_pdf(
10         qdf: Function, start_val: float = 0, center_on_zero: bool =
            False,
11     ) -> Function:
12         """Directly convert a qdf to a pdf using inverse function rule
            on cdf."""
13         if center_on_zero:
14             cdf = qdf.integrate().vcenter().invert()
15         else:
16             cdf = (qdf.integrate() + start_val).invert()

```

```
17         return 1 / qdf.compose(cdf)
```

B.2. Optimal Range Calculation in Python

This code shows our computation of the "effective" range by narrowing the domain of the pdf to the points where it is larger than $\delta = 10^{-3}$

```
1     def get_optimal_range(funcs: list[Function], delta: float = 1e-3) ->
        np.ndarray:
2         """Get narrower support if density values are too small (smaller
            than delta).
3
4         This is used so the qdfs dont get astronomically large at the
            boundaries and destroy
5         numerical methods.
6
7         Note: The method here assumes that the functions do have a
            compact
8         support (even if it is narrower than the initial support). So if
            there is a point x1
9         where func.y > delta is true, another point x2 > x1 where it is
            not, and then
10        another x3 > x2 where it is true again, then x2 is included in
            the new range
11        although it does not fullfill the condition.
12
13        """
14        new_ranges = np.zeros((len(funcs), 2))
15        for i, func in enumerate(funcs):
16            support_to_keep = func.x[func.y > delta]
17            new_ranges[i] = (support_to_keep[0], support_to_keep[-1])
18        return new_ranges
```

B.3. LQD Transformations in Python

The following code shows our Python implementation of the LQD transformation and its inverse.

```

1     def log_qd_transform(
2         densities_sample: list[Function], different_supports: bool =
           False,
3     ) -> list[Function]:
4         """Perform log quantile density transformation on a density
           sample."""
5         qdfs = [
6             pdf_to_qdf(density.drop_inf(), different_supports)
7             for density in densities_sample
8         ]
9         if different_supports:
10            qdfs_and_start_vals = np.array(qdfs)
11            lqdfs = [qdf.log() for qdf in qdfs_and_start_vals[:, 0]]
12            start_vals = np.array(qdfs_and_start_vals[:, 1], dtype=np.
              float64)
13            return lqdfs, start_vals
14        else:
15            return [qdf.log() for qdf in qdfs]
16
17    def inverse_log_qd_transform(
18        transformed_funcs: list[Function], start_of_support: list[float]
19        = None,
20    ) -> list[Function]:
21        """Transform back into density space."""
22        natural_qfs = [func.exp().integrate() for func in
          transformed_funcs]
23        if start_of_support is not None:
24            natural_qfs += start_of_support
25            cdfs = [qf.invert() for qf in natural_qfs]
26        else:
27            cdfs = [qf.vcenter().invert() for qf in natural_qfs]
28        exponents = [
29            -func.compose(cdf) for func, cdf in zip(transformed_funcs,
              cdfs, strict=True)
30        ]
31        return [exponent.exp() for exponent in exponents]

```

B.4. Solving Fréchet Quadratic Programming Problem

The following Python code shows the implementation of the quadratic programming problem to solve for the Fréchet means.

```

1      def solve_frechet_qp(
2          xs_to_predict: np.ndarray,
3          x_observed: np.ndarray,
4          quantile_functions: list[Function],
5      ) -> list[Function]:
6          """Sets up quadratic programming problem and solves it."""
7          estimates = []
8          for x in xs_to_predict:
9              # Estimate conditional qf, drop support where values become
10                 nan or inf
11              estimated_qf = qf_tilde(x, x_observed, quantile_functions).
12                 drop_inf()
13              constraints_check = estimated_qf.y[1:] - estimated_qf.y[:-1]
14              if np.all(constraints_check > 0):
15                  # If estimator valid qf, it is the optimal solution
16                  estimates.append(estimated_qf)
17              else:
18                  # Else, find closest vector to estimator that is a valid
19                     solution
20                  qp_a = estimated_qf.y
21                  grid_size = len(qp_a)
22                  qp_g = np.identity(grid_size)
23                  qp_c = np.eye(grid_size, grid_size - 1, k=-1) - np.eye(
24                     grid_size, grid_size - 1,
25                 )
26                  qp_b = np.zeros(grid_size - 1)
27                  solution = quadprog.solve_qp(qp_g, qp_a, qp_c, qp_b)[0]
28                  estimates.append(Function(estimated_qf.x, solution))
29          return estimates

```

C. Additional Figures

This chapter contains additional figures.

C.1. Illustration: Naive approach vs LQD approach

C.1.1. Data

We generated data as in the first simulation scenario in Petersen and Müller (2016): We sample 200 densities from truncated normal distributions, with $\mu = 0$ in the interval $[-3, 3]$. The sigmas are sampled as follows: we sample $\log(\sigma) \sim U(-1, 1)$ 200 times, and then obtain σ by calculating $\exp(\log(\sigma))$. Thus we have a sample $f_1, \dots, f_{200}$ of densities, with f_i being some truncated normal distribution with varying variances, centered on zero. The true center of this DGP is a truncated standard normal distribution, so that's what we would expect to see as estimated mean.

As a measure of explained variance, we used the Fréchet variance explained (fve) that Petersen and Müller (2016) defined. We refer to that paper for a definition of fve.

We used this measure in two ways: first, to judge the number K^* of how many components need to be included in the Karhunen-Loève representation to reach a given level of variance explained; and second, to compare the accuracy of the LQD method in the scenario with additional density estimation step to the one without this step. For this, $K = 1$ was fixed as this is the true number of principal components driving the variation in the sample curves (all variation in the sample is given by the variation in one parameter σ). Then the fve was calculated on the basis of the resultant Karhunen-Loève representations. More details are in Petersen and Müller (2016).

Before we present the simulation results, we turn to another visual comparison of the naive application of fda methods with the LQD method.

C.1.2. Modes of Variation

Here, we briefly show the difference in the estimated modes of variance from the naive and LQD approach. Figure 15 and Figure 16 show the first two modes of variation yielded by the two methods. One can see that the naive method yields very similar modes, explaining similar areas of variance while the LQD method captures different aspects.

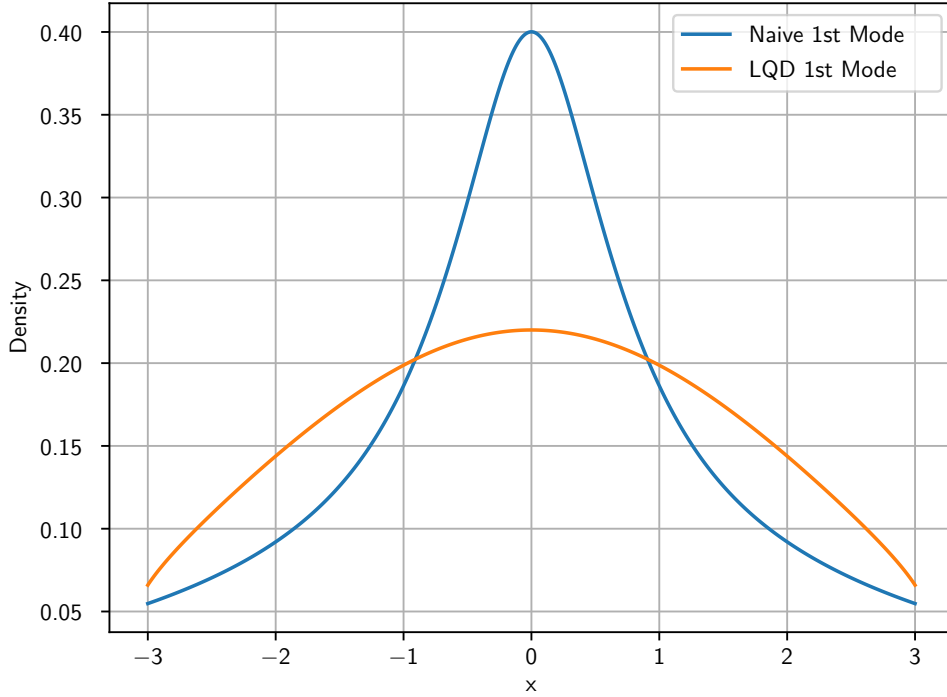


Figure 15: Comparison of first mode of variation of naive approach and of LQD approach.

C.1.3. Simulation results

The density estimation procedure is the same as explained in Section 7.2 but with one deviation: here, we used the boundary corrected estimator defined in Petersen and Müller (2016). Since we used the standard normal kernel, the estimator as defined in this paper, was not continuous. So we made it continuous by setting the weight function equal to

$$\left(\int_{-1}^1 \kappa(u) du \right)^{-1},$$

instead of 1 for values of x that are *not* near the boundaries.

Figure 17 shows the Wasserstein mean of the density functions, both in the scenarios with and without previous density estimation. Figure 18 shows the cross-sectional mean of the density functions, both in the scenarios with and without previous density estimation.

Figure 19 compares the Wasserstein mean to the cross sectional mean, in the scenario without density estimation and for $n = 200$.

Figure 20 shows the Fréchet

(Optimal K for directly observed densities always 1, so graphic omitted.)

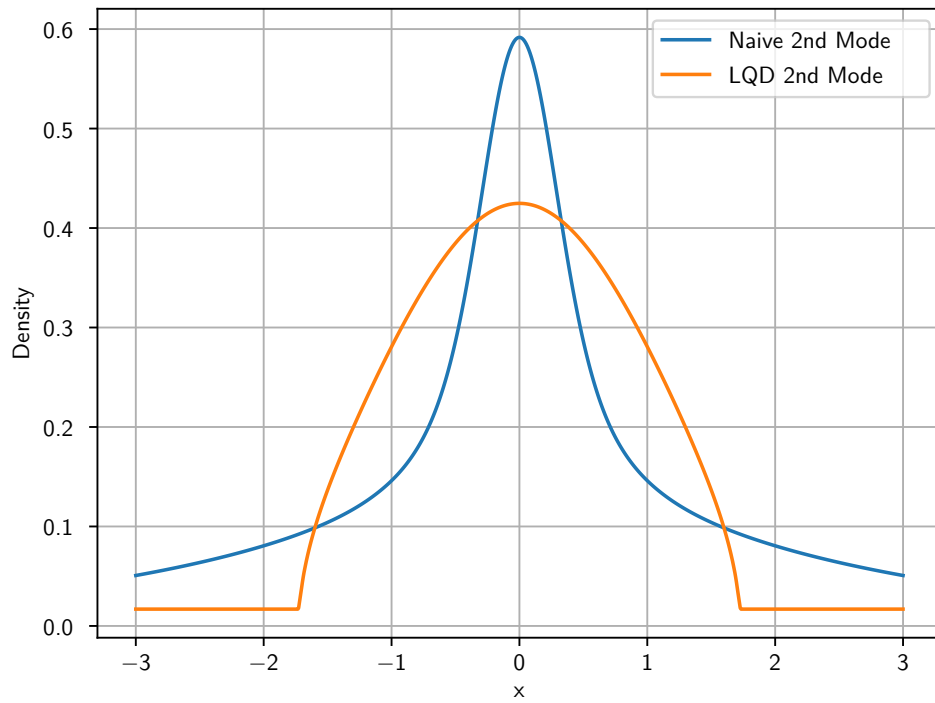


Figure 16: Comparison of second mode of variation of naive approach and of LQD approach.

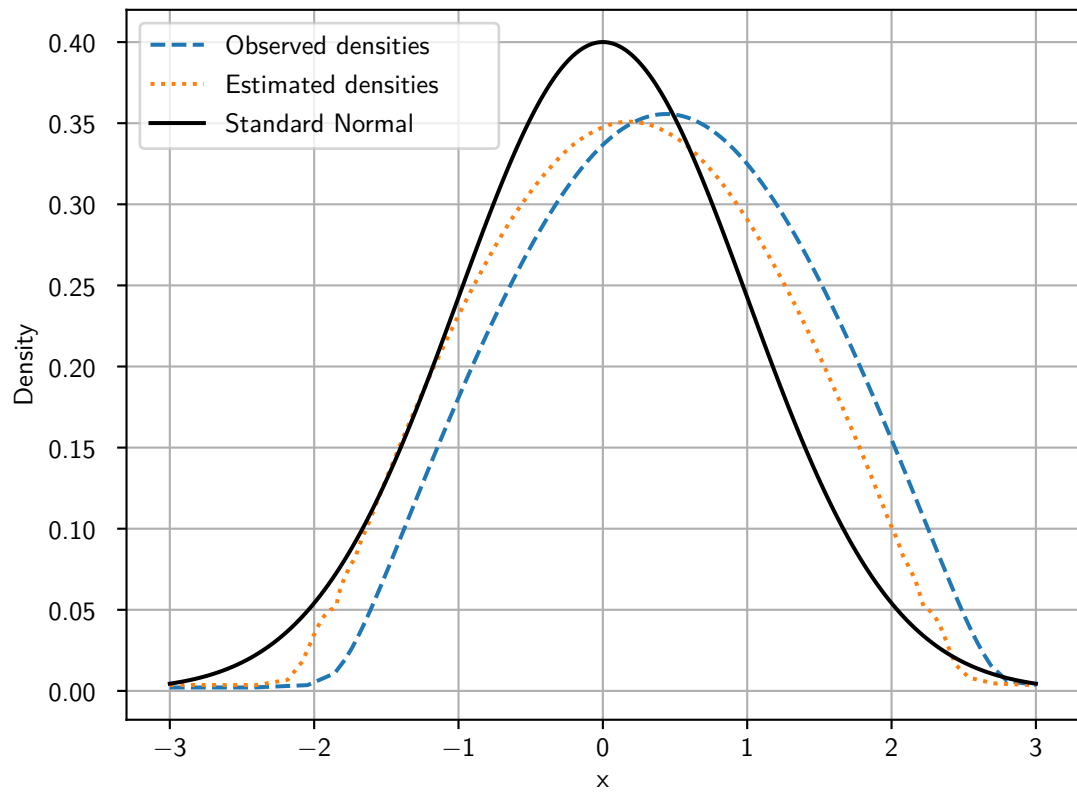


Figure 17: Simulation results of Wasserstein mean of observed densities vs. Wasserstein mean of estimated densities, $n = 200$.

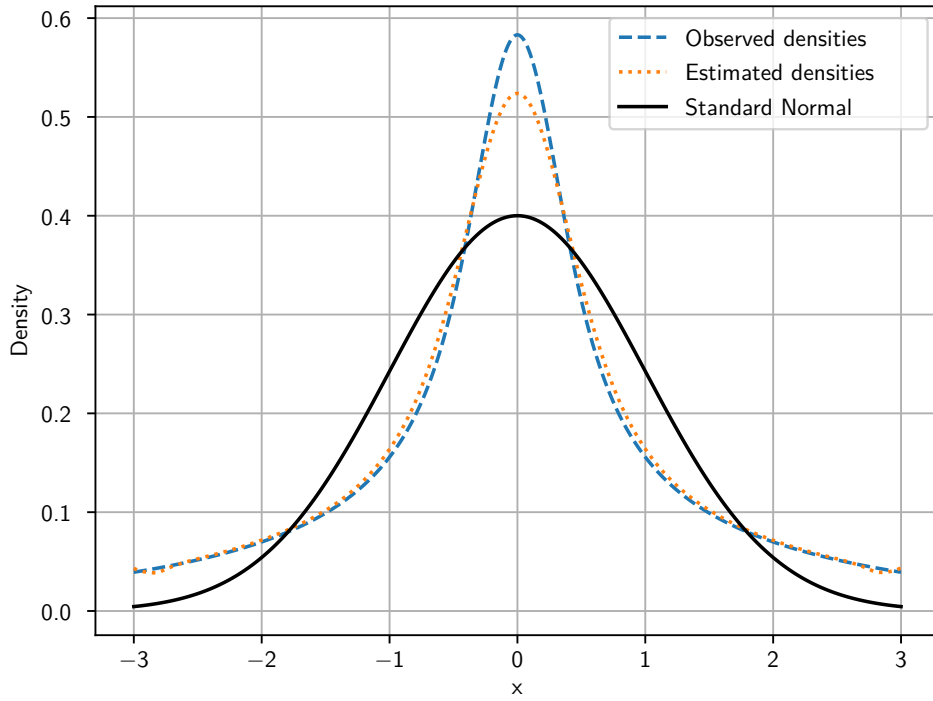


Figure 18: Simulation results of cross-sectional mean of observed densities vs. cross-sectional mean of estimated densities, $n = 200$.

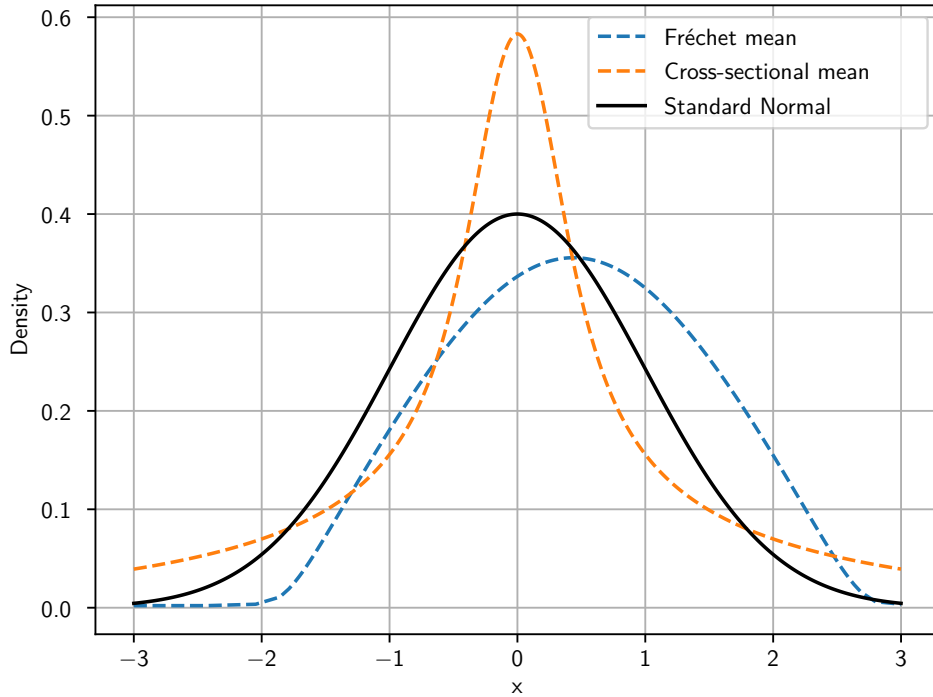


Figure 19: Simulation results of Wasserstein mean plotted against "naive" cross-sectional mean for $n = 200$.

C. Additional Figures

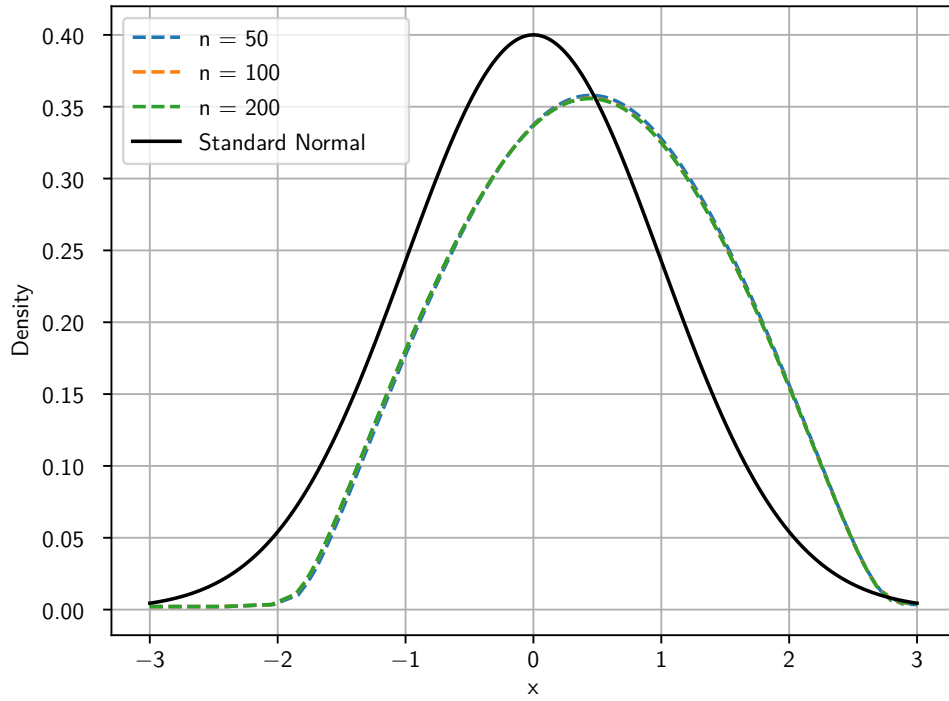


Figure 20: Simulation results for calculating Wasserstein mean densities with different sample sizes.

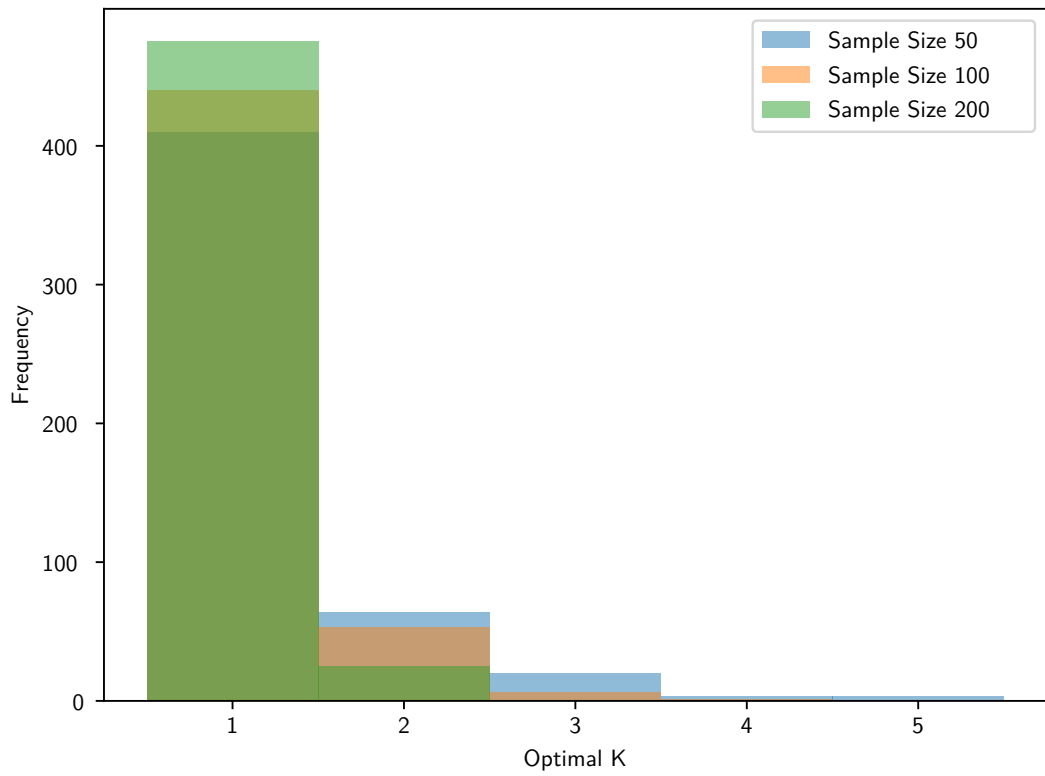


Figure 21: Simulation results for optimal choice of K in context of density estimation, given that $\text{fve} \geq 90\%$

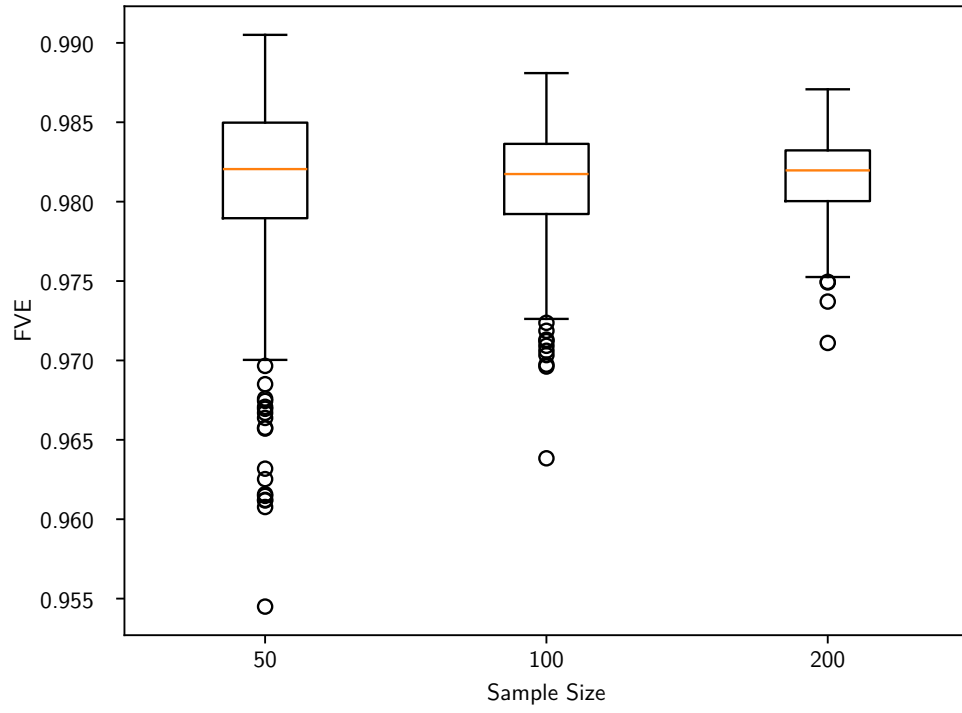


Figure 22: Boxplot of Fréchet fraction of variance explained with directly observed densities, when $K = 1$.

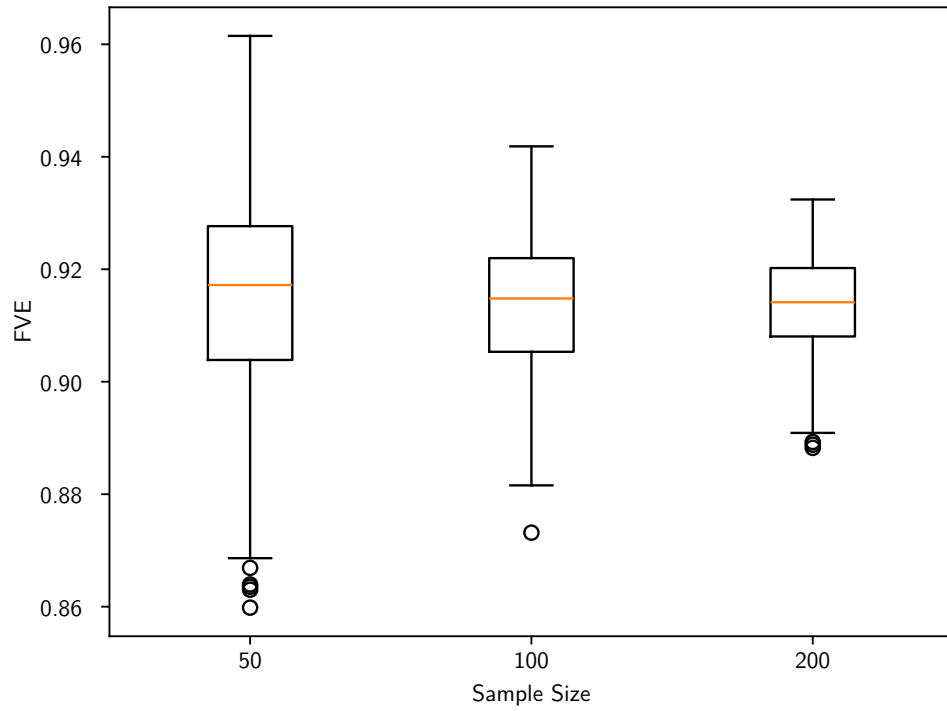


Figure 23: Boxplot of Fréchet fraction of variance explained with previously estimated densities, when $K = 1$.

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Date: _____

Signature: _____